

Unlocking the Data Ecosystem for African Startups

Mapping the innovations and the regulatory implications for data and AI-driven AgTech ecosystems across Africa

I. About



DATA
GOVERNANCE
IN AFRICA

Data Governance in Africa The Data Governance in Africa Initiative supports the African Union (AU) and its member states to form development-oriented and human-centric data regulation, data use and data infrastructure at continental, regional and member states levels. The Initiative provides concrete support for a single digital market in Africa by harmonising regulations on data, creating cross-border data flow and promoting investments in secure and sustainable data infrastructure facilities.

The initiative started implementation in 2023 with an investment of €60 million from the European Union, Germany, Finland, Belgium, France, and Estonia. It is jointly implemented by Enabel, Digital Africa, GIZ, ESTDEV, Expertise France and HAUS.



Digital Africa Digital Africa is a subsidiary of Proparco (AFD Group) dedicated to supporting the growth of Africa's tech ecosystems. It invests directly in innovative startups, with a particular focus on ecosystems underserved by traditional investors, and works to strengthen the enabling environment for digital entrepreneurship across the continent.

Through its programmes and partnerships, Digital Africa aims to equip African entrepreneurs with the tools, networks and capital needed to scale impactful digital solutions that respond to concrete societal and economic challenges. The AI 4 Startups (AI4SU) is a program implemented by Digital Africa in collaboration with Expertise France as part of the Data Governance in Africa initiative funded by the European Initiative. The AI4Startups programme supports African startups in building responsible and scalable AI solutions by unlocking the value of their own data. Through structured data governance frameworks and open-source model development, the programme enables startups to innovate with trusted, well-managed datasets.



Briter is a research and market intelligence firm focused on delivering insights into emerging markets across Africa, Latin America, and Asia. Briter offers a data platform, Briter Intelligence, which provides access to comprehensive company, investor, and funding data via interactive dashboards and customisable insights. **AgBase** is Briter's sector-specific data and intelligence platform focused on agtech and foodtech ecosystems across emerging markets. It provides structured data on funders, deal flows, sector dynamics, and investment trends, offering a comprehensive view of capital activity and market development in the agricultural sector. AgBase is powered by Briter and Mercy Corps AgriFin and funded by the Gates Foundation and the Foreign, Commonwealth & Development Office (FCDO).

II. Abbreviations

Abbreviation	Full form
AI	Artificial intelligence
AfCFTA	African Continental Free Trade Area
AUDPF	African Union Data Policy Framework
B2B / B2G / B2C	Business-to-Business/Business-to-Government /Business-to-Consumer
DFI	Development Finance Institution
DPO	Data Protection Officer
ESO	Entrepreneur support organisation
ESG	Environmental, social, and governance
FAO	Food and Agriculture Organization
IP	Intellectual property
IoT	Internet of Things
KALRO	Kenya Agricultural and Livestock Research Organisation
LLM	Large language model
MSME	Micro, small, and medium enterprises
MNO	Mobile network operator
NLP	Natural language processing
ODPC	Office of the Data Protection Commissioner (Kenya)
R&D	Research and development
SLM	Small language model
SME	Small and medium enterprises
SSA	Sub-Saharan Africa
VC	Venture capital

III. Contributors

This report integrates insights from AgTechs, policy researchers, AI training tractioners, and investors operating across the focus countries, in order to capture contexts and voices from the ground. We would like to extend our sincere appreciation for the time and insights shared by the contributors.

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V. Utilising AI to read this report

You can read this report the traditional way or let AI do some of the heavy lifting. Think of this report as a dataset you can query, not just a static publication. Upload the report to your favourite AI assistant and ask questions about ecosystems, funding patterns, policy priorities, or opportunities for your organisation. AI tools can help you quickly extract insights, compare ecosystems, and identify opportunities relevant to your role.



Tip: When using AI tools, you can start your prompt by specifying your position as a reader and with the following instruction to obtain more precise answers: *"Using only the information contained in this report, provide a concise answer to the following question:"*

Ask AI these questions about the report

General Insights

- What is the size of the data and AI-driven AgTech sector across the eight countries studied?
- Who is funding the data and AI-driven AgTech landscape across these markets?
- What open agricultural datasets are available for AgTech (optionally in [country])?
- How do data and AI-driven AgTech ecosystems differ across the countries covered in the report?

Funders

- How should I differentiate my support or funding strategy depending on market maturity?
- How to access an AgTech' adapts its strategy to its ecosystem?
- As a [precise your role], who are the key ecosystem partners I should collaborate with?

Startups

- What are the regulations and policies that affect data-driven AgTech in [country]?
- What types of ESOs help AgTech grow in the countries covered in the report?
- What does investment readiness look like for data-driven AgTech?

Policymakers

- What are the key data gaps in the ecosystem that regulations could address in [country]?
- What should be the main policy priorities and incentives for better data access in [my country]?



Prefer to explore the report yourself? Follow the chapter guide.

Summary

AgTech Landscape

Support Ecosystem

Data and Regulations

Diagnostic

Recommendations

VI. Executive Summary: bridging the data gap

This report maps the landscape of data and AI-driven businesses, ecosystem enablers, and policy players in the AgTech sector across eight Sub-Saharan African markets: Senegal, Côte d'Ivoire, Kenya, Rwanda, Tanzania, Botswana, South Africa, and Ghana. The study identifies a foundational reliance on data for AgTech activation and proposes a sequenced, policy-driven roadmap to address structural challenges and unlock the ecosystem's potential.

The key findings from the study are outlined below.



A new wave of AgTech innovation is emerging: The data and AI-driven AgTech sector in Africa is growing, with one-third of all AgTechs adopting data and AI technologies. The sector's growing momentum is attracting investor interest, with innovators raising close to \$400M in disclosed funding in the last 10 years. However, this growth has largely been built upon **small, localised datasets** on farms, farmers, and agriculture value chain intermediaries. These datasets help AgTechs build highly localised solutions, but many struggle to scale due to the high cost of developing them. The larger, integrated national datasets using geospatial, national-level soil health, and real-time market prices are not available or are disaggregated for AgTechs to access efficiently.

The small and localised datasets reflect structural barriers:

- Data fragmentation and silos**
 Fundamental agriculture data, such as soil, weather, and land use data, is often fragmented, inaccessible, or missing. Data collected at the AgTech level is costly and becomes important proprietary data, so they shield their data from sharing. Public institutions also lack incentives for cross-institutional sharing, creating silos.
- High costs**
 The cost of data collection, storage, and processing is high and can be unpredictable. This creates a barrier to scaling, especially with limited localised capacity to compete with global pricing.
- Talent drain**
 A lack of financial incentives and a limited pipeline between training institutions and the market leads to brain drain of skilled AgTech and AI talent.

Bridging these gaps requires tailored strategies that align with the maturity of the AgTech ecosystem. Funders, support organisations, and policymakers all have a critical role to play in building the necessary data capabilities and enabling conditions for AgTechs to build momentum and new capabilities within data and AI. The report advocates for a sequenced approach by market, building capabilities first through digitalisation, data enablement, data harmonisation, enabling open data, and building AI capabilities through better data development.

In order to advise this sequential approach, the report introduces a diagnostic framework to assess the state of the ecosystem. The framework synthesises the countries' innovation ecosystem, macroeconomic environment, and the enabling environment to categorise them into four archetypes: **nascent, activation, market-responsive, and scaling ecosystems, which guide ecosystem growth priorities.** The table below introduces the high-level learnings by each archetype.

Maturity archetypes	Startups	Funders & ESOs	Policymakers
Nascent (Botswana, Namibia)	Build digital foundations before heavier data and AI-driven solutions.	Invest in basic technical readiness, more women and youth talent development	Lead state-led data collection
Activation (Senegal, Tanzania, and Côte d'Ivoire)	Building sound businesses that can collect high-quality data based on your user base	Develop more high-quality talent and retain them in the ecosystem while leveraging more concessional and patient capital to derisk the ecosystem	Enable open data, and making them available at a more consistent
Market-responsive (Ghana, Rwanda)	Based on the data developed from your user base, develop the most useful use cases through technology, including using voice-first AI capabilities	Increase visibility on success cases to attract a more diverse group of funders to the space	Build better data operability across government institutions
Scaling (Kenya, South Africa)	Find more 2B growth pathways and better data governance to scale	Finding more fit-for-purpose financing for different companies emerging in the ecosystem	Build better IP and data protection mechanisms to encourage more collaborative innovation

In addition to the country-level policy priorities, strategic interventions are needed across all markets:

● Lower the cost of data collection by implementing incentives such as lower import taxes on IoT devices or training more IoT field agents. ● Develop policy direction to mobilise the industry as policy directions on data protection and Intellectual Property (IP) laws to provide security for innovators and improve investor confidence. ● Mobilise the sector by developing B2G opportunities, as AgTechs and data companies specialising in data collection and AI tech development can scale further by entering into government contracts and the public sector ecosystem. ● Create more data-specialised training programmes to develop a better talent pipeline. Universities and training institutions need to update their technology and curriculum to enhance the capacity of a data-enabled talent pool. This talent pipeline also needs to be plugged into the industry better via early talent-to-industry matching programmes.	● Invest in talent-to-industry pipeline by developing data stewardship capabilities within universities, and creating more direct pathways into industry roles in collaboration with the EdTech sector, following models such as the T4SU program¹. ● Build use cases in partnerships with the private sector to test the data quality and data sharing mechanisms in an applied setting. Use cases can help shape the burgeoning policy landscape. ● Leverage the private sector as soundboards by developing industry advisory boards to test and assess the new policy landscape. Build advisory boards with active ecosystem actors (AgTechs, funders, ESOs, training EdTechs) to get insights on the efficacy of new data-related policies developed in recent years. ● Highlight the success cases to mobilise more funder interest, and to bring to light growth pathways and exit possibilities for data-driven AgTechs.
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Beyond data availability gaps, the study finds that one of the bottlenecks across markets lies in the lack of trusted governance mechanisms that enable structured, incentive-aligned data sharing across institutions, sectors, and borders. Opportunity enabling data governance, therefore, means designing frameworks that balance protection, interoperability, and economic value creation for startups. By combining ecosystem mapping with applied experimentation through AI4Startups, this report by Digital Africa contributes a startup-centred perspective to the broader Data Governance in Africa initiative.

1. The T4SU (Talent 4 Startups) program, implemented by Digital Africa, supports the training of young professionals in high-demand tech and digital skills across the African continent through a partnership-based approach aligned with startups' needs.

Introduction

In Africa, data and artificial intelligence (AI) are transforming agriculture by enabling smarter and more timely decisions across the entire value chain. From digital platforms and mobile advisory services to sensors, satellites, and predictive analytics, these technologies are boosting productivity, optimising resource use, improving weather and market forecasting, supporting early detection of risks, and driving greater sustainability and resilience.

Not only is AI changing the landscape of innovation, but its market potential is also immense. The African AI market is estimated to have taken 2.5% of the global AI market value in 2023². AI technologies are projected to add \$2.9 trillion to the African economy between 2024 and 2030, which is equivalent to the expected total GDP of African countries in 2025³. The ability to capture the opportunities in this growing market will become a crucial task for many African countries to bridge the digital and AI divide.

While data is increasingly recognised as a cornerstone of innovation across many sectors, most startups have limited access to updated, high-quality, and comprehensive data. In the agriculture sector, this limits the potential for innovators to address real farmer challenges. Despite rising digital adoption rates and the growing use of artificial intelligence (AI), many entrepreneurs also operate in fragmented environments shaped by regulatory uncertainty, uneven infrastructure, and weak data governance. These data gaps not only slow product development but also reduce the ability of startups to scale, validate AI models, and reach new customer segments.

The 'AI 4 Startups: Unlocking the Data Ecosystem' project has been launched in response to these challenges. The initiative seeks to map and analyse the ecosystem of data-related actors across eight African countries, and focuses on understanding the networks, capacities, and governance frameworks that shape how startups access, share, and use data, with a focus on the agritech sector.

The study aligns closely with the challenges addressed by the AI4Startups programme, which has served as a practical experimentation ground for some of these dynamics, notably through cross-border hackathons and structured data-sharing agreements. These applied use cases were designed as innovative approaches to enhance data sharing and value creation with and for startups. Through this bottom-up methodology, AI4Startups also helped surface concrete regulatory and data ecosystem barriers faced by innovators.

This report **aims to identify clearer pathways for data and AI-driven startups in Africa by mapping the landscape of current data ecosystem mechanisms and players addressing the foundational elements required for AI to work effectively**. The report also explores gender and inclusion dynamics within the data economy and provides recommendations and partnership opportunities to strengthen cross-border data collaboration and promote responsible, open data use for African innovation.

2. This is lower than African GDP compared to global GDP, which is 3%. Source: https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/wp-content/uploads/2024/07/AI_for_Africa.pdf

3. [https://africa.businessinsider.com/local/markets/africas-gdp-is-projected-to-increase-from-dollar23-trillion-in-2023-to-dollar28/9w-859wq#:~:text=Africa's%20GDP%20is%20projected%20to,\(IMF\)%20World%20Economic%20Outlook](https://africa.businessinsider.com/local/markets/africas-gdp-is-projected-to-increase-from-dollar23-trillion-in-2023-to-dollar28/9w-859wq#:~:text=Africa's%20GDP%20is%20projected%20to,(IMF)%20World%20Economic%20Outlook).

The report is structured as follows:

Chapter 1

Provides an overview of the data and AI-driven AgTech landscape in Africa and its funding across the last 10 years.

Chapter 2

Identifies the ecosystem stakeholders that support, fund, and fuel the data and AI-driven AgTech innovation.

Chapter 3

Explains the policy and regulatory environments across the 8 focus markets.

Chapter 4

Assesses the emerging archetypes of the 8 ecosystems using a systematic diagnosis framework.

Chapter 5

Offers recommendations for creating a stronger environment for data and AI-driven AgTech innovation ecosystems in Sub-Saharan Africa.

To assist in navigating the report, we have integrated stakeholder-specific highlights throughout the chapters. These action notes, marked with the visual cues below, are tailored to the distinct priorities of three key groups: agtechs, funders and support organisations, and policymakers.



Startups



Funders



Policymakers

Methodology

The quantitative data draws on Briter Intelligence datasets, focusing on eight key markets in Africa (Senegal, Côte d'Ivoire, Kenya, Rwanda, Tanzania, Botswana, South Africa, and Ghana) that form part of the AI 4 Startups project focus countries. Across these markets, Briter data covers 1700+ funded startups, and 250+ AgTechs and their disclosed funding in the last decade. Amongst the AgTech dealflow, 140 funded AgTechs were identified as AI and data-driven AgTechs, and they have altogether raised \$400M in disclosed funding through 69 disclosed deals with known funding size. In addition, there are 30+ disclosed deals without known funding size that are included in the counting of deal flows but excluded from analysis on deal size. Briter Intelligence data also features 140+ ESOs and 250+ funders working with AgTechs in the select markets. In the research, additional data mapping 70+ policies on data and AI development, 130+ universities and research institutions, and 30+ regulatory institutions engaged in enabling better AI and data technology development. Information about policy, regulatory institutions, and research/university organisations is collected directly from the institution's self-reported information and desk research. Additional desk research leveraged open data from credible sources such as the World Bank, FAO, ILO, and Open Data Watch.

The core qualitative insights feeding into this report were gathered through 10+ direct stakeholder engagements with investors, policy researchers, AgTechs, and ecosystem builders.

Chapter 1: Innovation and funding landscape

The landscape of data and AI-driven AgTech startups is growing across Africa. This chapter analyses the growth through indicators such as startup density, sectoral distribution, and the evolution of the funding landscape. The scope of the landscape analysis is on eight key markets: Senegal, Côte d'Ivoire, Kenya, Rwanda, Tanzania, Botswana, South Africa, and Ghana. These countries were selected to represent different geographical zones and varying degrees of market conditions across the continent. Additionally, there are existing and ongoing data governance initiatives by GIZ and the AU, and hence present synergetic opportunities to apply the learnings from this report.

1.1. Defining data and AI-driven AgTechs

This research defines AgTechs as startups and high-growth SMEs that produce products and services with innovative digital and technological components in the agriculture sector. The term “data and AI-driven” indicates that an AgTech’s core product offerings are anchored on the availability of data and AI technologies and that AI technology and data are central to its value proposition.

This report categorises AgTechs by their technological adoption levels, as defined below. The analysis of data and AI-driven AgTechs throughout the report includes AgTechs with mid and high levels of adoption in data and AI.

	Data-driven AgTechs	AI-driven AgTechs
High adoption	AgTechs that have data as a core part of their product and service, such as data-as-a-service. Example: Farmerline has built a data platform called Mergdata that helps agribusinesses, NGOs, and governments drive farmer profitability.	AgTechs that have AI as a core part of the product or are building AI-native tools. Example: Rhea has built an AI model to provide predictive soil health advisory services
Mid adoption	AgTechs that use data to improve their products, services, and operations. Example: mAgri integrates weather data to their marketing platform to add value to its platform functions.	AgTechs that use AI as an enabler to their product, service, and operations. Example: Almacena integrates AI into its finance and digital supply chain solutions to help commodity agribusinesses grow and operate more efficiently.

African AgTechs have been rising both in numbers and scale in the last ten years, developing technologies and services that solve real-life challenges that farmers face on the continent. AgTechs range from pre-farm activities such as biotech for seed and online marketplaces for inputs, to on-farm activities such as precision farming technology, farm management software, financing for farmers that were previously underserved by traditional lending institutions, and data-driven crop insurance, as well as to off-farm activities such as real-time market price information, solar-powered cold storage that reduces post-harvest loss, and apps that connect fragmented logistics chains. Digitalisation of the agriculture sector is contributing to formalising the agriculture sector in Sub-Saharan Africa (SSA)⁴, and promoting more growth and efficiency⁵ in one of the biggest sectors on the continent.

Figure 1: Data and AI-driven AgTechs across 8 African markets



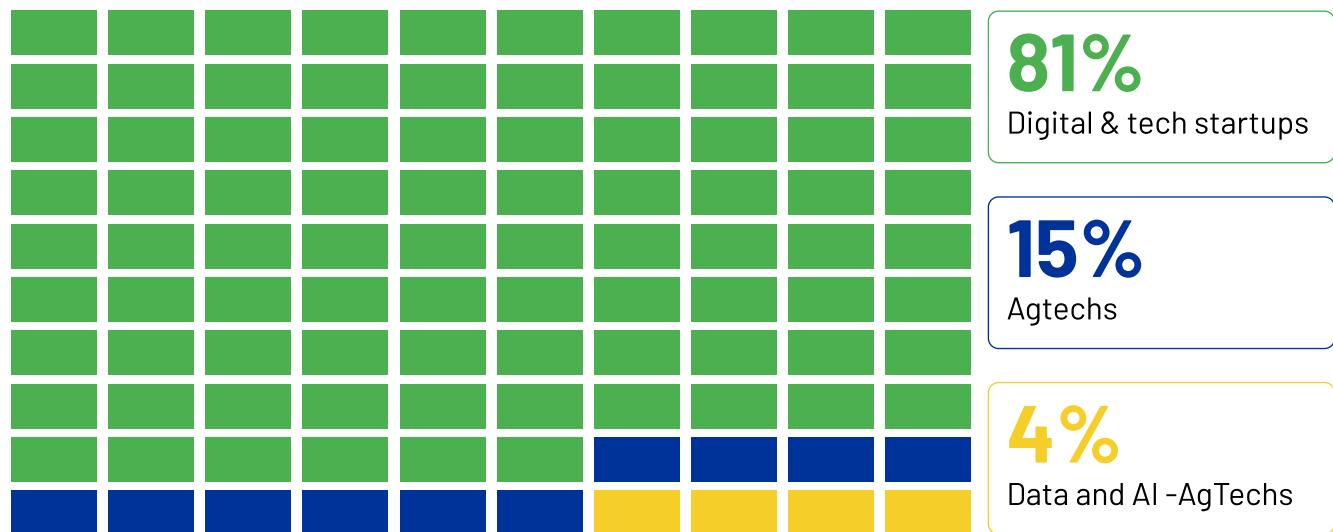
Source: Briter Intelligence and AgBase (2026)

A third of the AgTech sector is data and AI-driven. Among the 450+ AgTech identified in the ecosystems of the selected eight study countries, a quarter of AgTechs are identified as data and AI-driven as they demonstrate high or mid-adoption of data and AI capabilities. The data and AI-driven AgTechs identified take up 4% of the broader tech startup ecosystem across the eight countries. While data and AI-driven AgTechs are proportionately small compared to the broader innovation ecosystem, they have raised over \$350 million in the last 10 years, constituting about 2.5% of the overall funding landscape.

4. <https://www.sciencedirect.com/science/article/abs/pii/S0743016725000774>

5. <https://alliancebioversityciat.org/stories/digital-agriculture-boosts-crop-yields-efficiency>

Figure 2: Size of the market



Source: Briter Intelligence and AgBase (2026)

Note: Market size by the overall tech startup ecosystem, AgTech, and data and AI-driven AgTechs



Startup note: Sequencing innovation

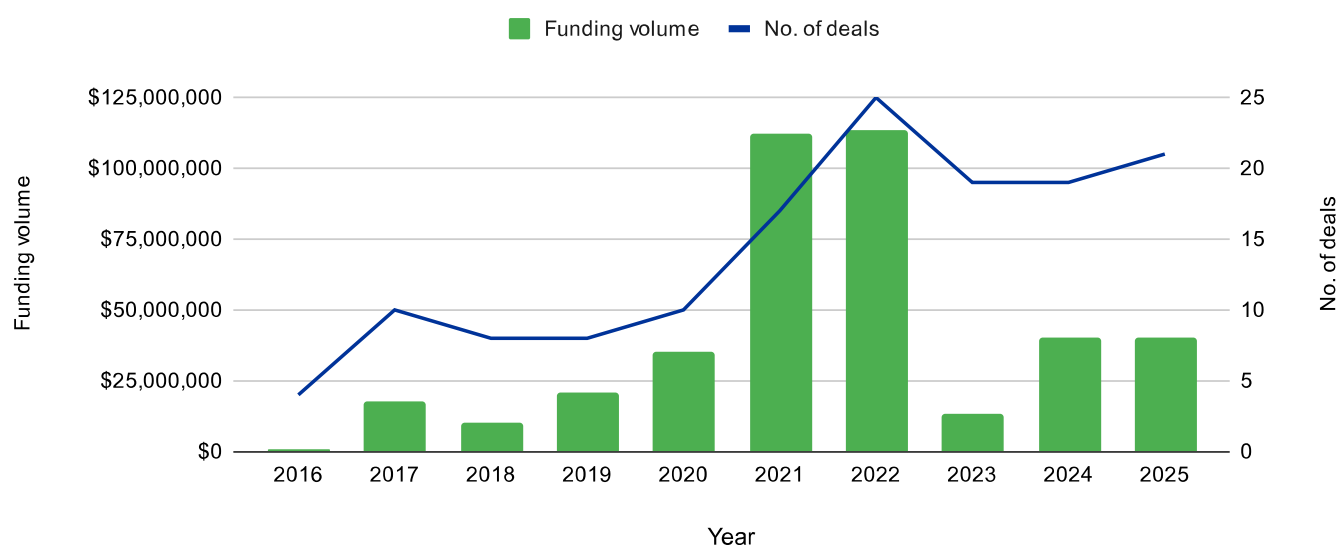
Innovation can be sequenced. AgTechs can benefit from focusing on establishing a digital and data-driven foundation before developing more complex models. Focus first on the digital tools (apps, registries, sensors) that capture information to create the high-quality data "fuel" you need for AI later.

Data and AI offerings are tightly related to each other, especially as a response to the lack of open data in the market. More than half of the data and AI-driven AgTechs studied adopt both data and AI practices within their product or service offerings. On one hand, this reflects the order of AI technology development, where high-quality data can help develop high-quality AI models. On the other hand, it highlights one of the market challenges prevalent in the African context – the lack of open data on agriculture. As Warren Le Fleur from Koya Capital states, "data is like gold in Africa." The lack of open-source data on agriculture or market information pushes AgTechs to adapt by developing their own proprietary datasets. This means many AgTech companies start off by maximising consumer data capture during the adoption of their flagship products, and then develop AI-enabled tools that utilise the data collected from the existing consumer base. For example, Rhea Africa collects soil health information from their consumers and then uses an AI system to aggregate data and analyse it before generating a report for their consumers.

1.2. Funding landscape

Funding for data and AI-driven AgTechs has grown over the last 10 years, raising close to \$400M across the eight countries. This growth trend follows the broader AgTech sector. The growth trend peaks in 2021/22, and stabilises from 2022 onwards. The main difference between data and AI-driven AgTechs and the broader AgTech funding landscape (see [Annexe](#)) is that 2025 did not show a large dip in the funding size. This reflects an increased interest in the intersection of data and AI and the AgTech sector.

Figure 3: Funding flow to data and AI-driven AgTechs



Source: Briter Intelligence and AgBase (2026)

Larger markets such as Kenya and South Africa lead in terms of the number of funded data and AI-driven AgTech solutions and the volume of mobilised funding into AgTechs. While larger markets naturally have a greater number of innovations and capital availability in absolute terms, when looking at the proportionate market size, Kenyan and South African AgTech sectors are proportionately small in the overall innovation ecosystem (17% and 11% of overall funding size, respectively), and more VC-driven sectors such as FinTech and e-Commerce dominate. On the other hand, medium-sized markets such as Ghana and Tanzania have proportionately larger AgTech sectors compared to the overall innovation ecosystem (30% and 35% of overall funding size, respectively), indicating that AgTech sectors, while smaller compared to Kenya and South Africa, are attracting founders and funders. Rwanda, Senegal, Botswana, and the Côte d'Ivoire showed fewer (less than 10) active data and AI-driven AgTech solutions, and signals that the overall AgTech sector is still at an early stage of development.

While software solutions attract more funding, there is an increased interest in hardware solutions. Many data and AI-driven companies develop software solutions that are asset-light and scalable, and this has proven to be more favourable for venture capital investment. Digital Green's FarmerChat, which provides AI farm advisory services over chat apps, is a prime example of a software-based AI AgTech solution. However, given the need to capture more data for precision agriculture or deeper AI tech development, there is an increased interest in funding and support

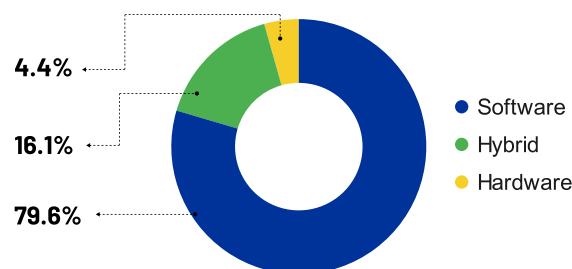
for hardware technologies such as smart solar-powered cold chain solutions, and hybrid solutions such as smart irrigation tools that are connected with AI-powered software. This integration is often a response to the scarcity of open-source data, compelling startups to prioritise developing their own high-quality proprietary datasets, which in turn fuels the development of specialised AI products.

Women founders are visible in the earlier rounds, even more than in other sectors, but a limited number have raised in the growth stage.

While men founders make up the majority of the funded data and AI-driven AgTechs, women founders are fundraising at a higher proportion than the ecosystem average. 20% of funded data and AI-driven AgTechs are founded by women, which is higher than the overall AgTech funding landscape where only 17% of funded AgTechs are founded by women, or the overall African innovation ecosystem where only 12% of funded companies are founded by women. Women founders are mostly raising grants and early-stage rounds through competition and accelerators compared to AgTechs founded by men. This is reflected in the median ticket size difference between the two groups where the median ticket size for women-founded AgTech is \$100,000 compared to \$950,000 for mixed-gender teams and \$1.5 million for AgTechs founded by men.

The fundraising landscape shows much data, and AI-driven AgTechs are raising at an earlier stage, with several raising into the growth stage. 22% of all funding flows are deals under \$50,000 ticket sizes, which are mostly grants or in-kind assistance through incubator, accelerator, and pitch competitions, reflecting the ongoing ecosystem activation efforts. 30% of overall funding is below \$1 million ticket sizes, indicating funding at pre-seed and seed stage, and supporting AgTechs that are proving their business models and product-market fit. This evidence of growth-stage data and AI-driven AgTechs that have successfully raised between \$1 million and \$20 million ticket sizes, but 85% of growth stage fundraising happened in maturing markets such as Kenya and South Africa, with some activity in Ghana.

Figure 4. Software, hardware and hybrid solutions in data and AI-driven AgTechs



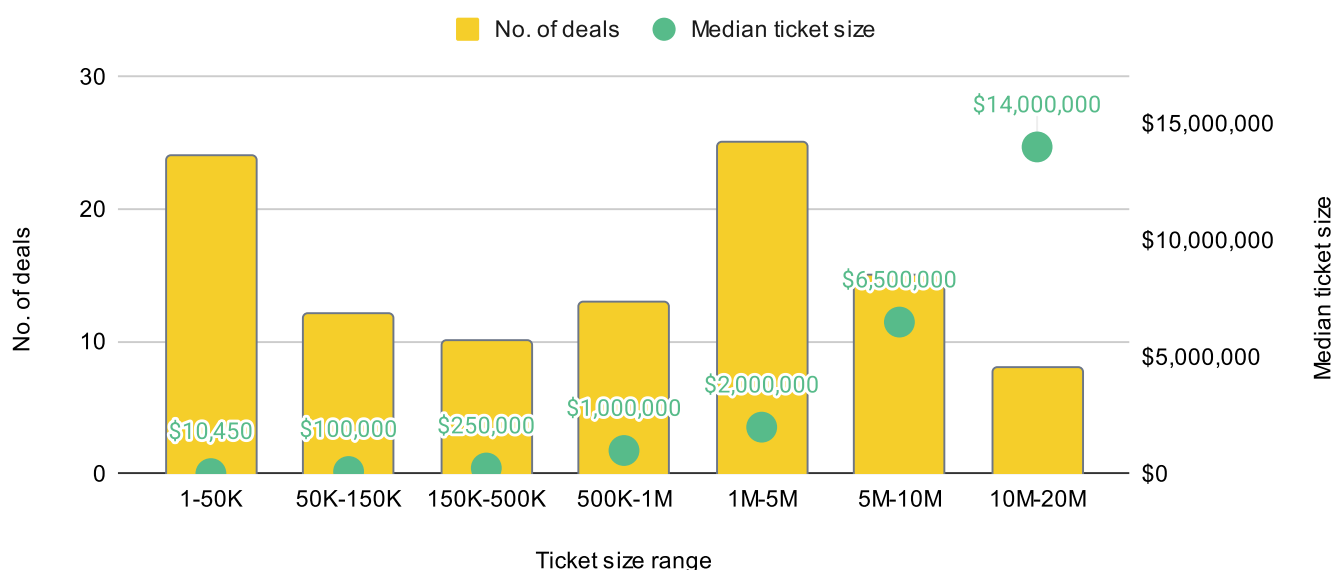
Source: Briter Intelligence and AgBase (2026)

Figure 5: Gender Analysis



Source: Briter Intelligence and AgBase (2026)

Figure 6: Volume of funding and median funding size by ticket size range

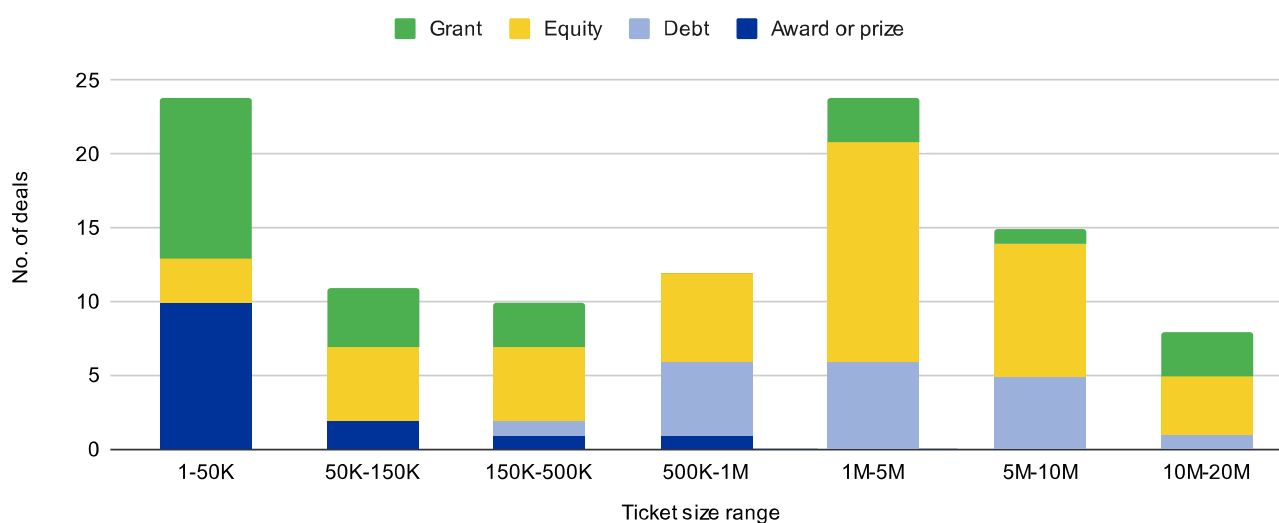


Source: Briter Intelligence and AgBase (2026)

Note: A quarter of deals (n=39) do not have a disclosed funding amount and are excluded from this analysis.

Grant and prize funding dominate very early stage working capital financing, and continue to supplement the funding landscape along with private capital at later stages. Equity financing is also widely used for pre-seed and seed stage, indicating data and AI-driven AgTechs are demonstrating commercialisation pathways that are meeting investor appetites. Debt financing is limited, and mainly comes from DFIs rather than local banks, as data and AI-driven AgTechs struggle to access debt from banks. DFI-backed debt funding has gone to a few AgTechs that demonstrate a sizeable customer base feeding into their data-driven service and a sound business model, such as One Acre Fund, Ignitia, Apollo Agriculture, mAgro, and Farmerline. Similarly, a few scaled AgTechs such as Pula and One Acre Fund have mobilised grant funding with \$1 million or higher ticket sizes from private foundations and the bilateral DFIs.

Figure 7: Funding instrument by number of deals and funding size



Source: Briter Intelligence and AgBase (2026)

Note: A quarter of deals (n=39) do not have a disclosed funding amount and are excluded from this analysis.

Chapter 2: Stakeholders supporting the data and AI-driven AgTech ecosystem

Most innovators can not scale their solutions by themselves. Stakeholders across the private and public sector operate in the AgTech and the broader innovation ecosystem to facilitate access to capital, specialised training, and strategic networks. This chapter explores the infrastructure of support available to data and AI-driven AgTech innovators across the eight focus countries. The support ecosystem mapping includes an overview of funders, entrepreneur support organisations (ESOs), and training and research institutions.

2.1. Funders landscape

More than 100 funders have funded data and AI-driven AgTechs across the eight focus countries in the last decade, but 70% are headquartered outside of Africa, mainly in Europe and the US. When South Africa is taken out, the proportion of local investors drops even further. This higher engagement of foreign funding reflects the funder landscape in the broader African innovation ecosystem, and is not specific to the AgTech sector. Kenya, one of the more mature ecosystems, has a large ecosystem of foreign funders, including DFIs, impact investors, and other international organisations. Hence, international funders make up 85% of the funding ecosystem, while 60% of funders are international funders in South Africa. Compared to South Africa and Kenya, local investors in the rest of the focus markets make up 10% or less.

The map below shows African and non-African funders that have funded data and AI-driven AgTechs across the eight focus markets in the last 10 years.

Figure 8: Funders of data and AI-driven AgTechs

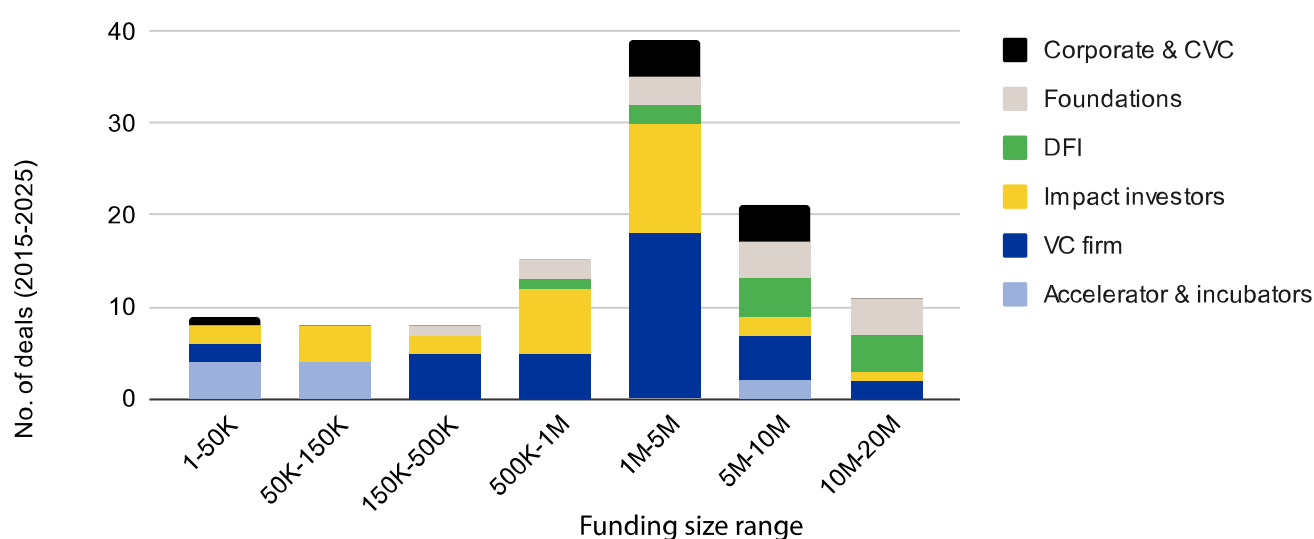


Source: Briter Intelligence and AgBase (2026)

Note: This map showcases funders who have participated in one or more funding rounds for African data and AI-driven AgTechs.

The funders come from diverse sources, including private financing, concessional impact investing, development financing, and corporate financing. Incubators and accelerators serve as the first providers of working capital to data and AI-driven AgTechs, as they provide in-kind and cash support ranging between \$1K to \$150K for their cohort of startups. VC and impact investors are the largest group of funders for AgTechs that manage to raise funding between the \$150K and \$1M ticket size range at the pre-seed to Series A funding. Development and corporate funders begin to back early-stage AgTechs from a \$1M ticket size, and become more engaged as companies fundraise at the growth stage, backing them at the \$5M-20M ticket size range. Corporates, especially in more mature markets, acquire high-performing data and AI-driven AgTechs, shown through Execo's acquisition of AfricaAI and Maximeyes Group's acquisition of Bamba Corp.

Figure 9: Funder types by funding size and no. of deals



Source: Briter Intelligence and AgBase (2026)



Funder note: Navigating the instrument matrix

DFIs, Impact VCs, and bilateral donors have differing risk appetites. Funders interested in entering the space should explore finance structures that match the instrument to the startup's stage and product type, for instance, utilising grants for early R&D of more AI-driven models for de-risking and reserving equity or debt for when unit economics are proven.

2.2. Ecosystem support organisation (ESO) landscape

Over 140 ESOs work with startups, youths, and small businesses to equip them with better data and AI capabilities across the eight focus countries. The map below shows the accelerators, incubators, innovation hubs, grassroots communities, and competitions that activate and accelerate data and AI-driven AgTechs. The ESO landscape map below reflects the difference in ecosystem enablement capacities across focus countries. Kenya and South Africa show a deep network of ESOs, and some are now developing more vertical expertise. Amongst more emerging countries, Ghana and Rwanda have activated more early-stage support and enablement initiatives compared to Senegal, Côte d'Ivoire and Tanzania.

Figure 10: ESOs map

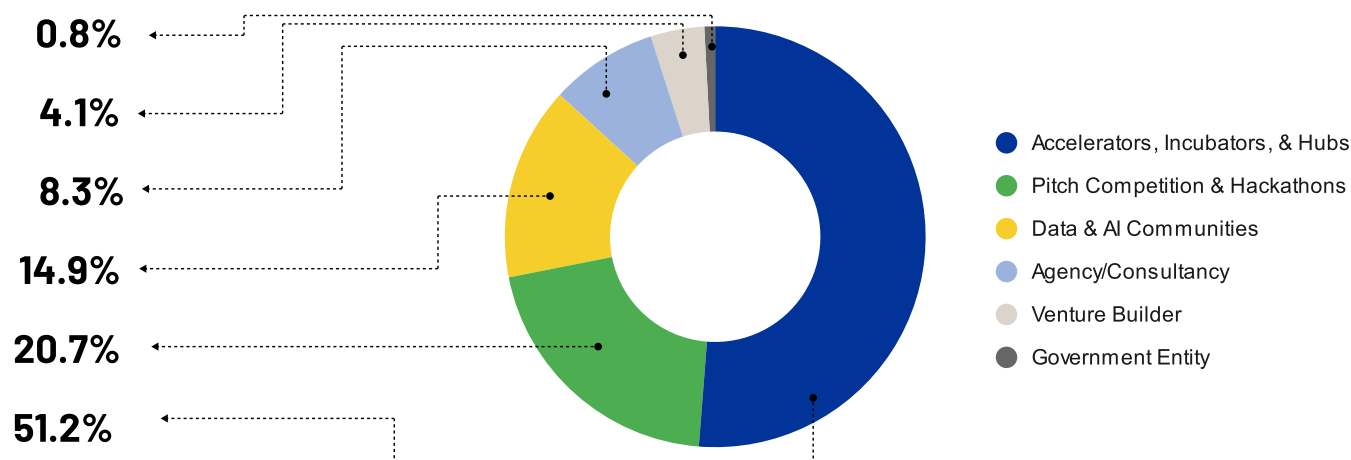


Source: Briter Intelligence and AgBase (2026)

Note: ESOs featured on the map support data and AI-driven AgTechs and/or offer specialised programmes, research, innovation spaces, resources, and networks to enable AgTechs to build data and AI-driven products or integration.

There is a rise of ESOs and communities specialising in the broader data and AI-driven ecosystem activation, even if the cross-sectional support between data, AI, and AgTech is limited. **One-third of these ESOs are incubators, accelerators, and innovation hubs** (including maker hubs), such as the pan-African BRAIN accelerator by Open Startup, Disraptor in South Africa, AI Hub Senegal, UNDP's AI Hub for Development, and Strathmore University's Agri-Food Innovation Center in Kenya, which provides hands-on support for early-stage companies to develop better data and AI capabilities.

Figure 11: ESO by type



Source: Briter Intelligence and AgBase (2026)

Another rising support type for data and AI-driven talent is pitch competitions, hackathons, and datathons. These events are often thematically focused and typically made accessible via an online format to bring together talents to develop a specific use case around a shared resource. This concentrated format helps shorten the R&D cycle. Examples of these are Digital Africa's online Datathons with Zindi, BIUST Machine Learning & AI Hackathon in Botswana, Ghana AI Hackathon, and AI Connect Hackathon in Rwanda.

Data and AI-specialised communities are creating more ground-up initiatives to bring together experts, students, and talents through organised events or online forums. They hold workshops, hackathons, or datathons, and develop collaborative projects that foster training and learning in the ecosystem. While they often run through grassroots and community-first approaches, some, such as Global AI Accra and AI Saturdays in Rwanda, are a part of a global network. These ground-up communities are designed to create an organic network of talents, students, and experts.

One additional group of AI-enabling actors are **AI consultancy services** that play an important role in capacitating companies and institutions that are not yet data-driven in integrating and building better data infrastructure. These companies may not be as visible as data and AI-driven AgTechs, but these SaaS and operations-driven companies serve as an important connectivity tissue for many of the markets where economies are transitioning from digital to data-driven economies. Examples of these companies are Enoveta in Rwanda or the African Academy of Artificial Intelligence (AAAI) of South Africa.

2.3. Universities, training, and research institutions

Universities, training, and research institutions as a whole supply human capital and contextual knowledge to the overall ecosystem. Universities and training institutions develop the talent pipeline in the agronomics, data, DevOps, and AI fields needed for the growing industry. Research institutions develop new insights or serve as knowledge aggregators in the agriculture sector, or develop new cutting-edge data-driven studies that can change how AgTechs develop their solutions. This chapter maps the universities, training facilities, and research institutions playing a key role in agriculture or data across the eight focus countries.

Figure 12: Universities, training, and research institutions map



Source: Briter Intelligence and AgBase (2026)

Over 130 institutions across the eight study countries provide training for data and AI skills. Half of these training facilities are public universities, **public research institutions, and technical institutes** that provide skills development via formal degrees or training modules.

After universities, EdTechs are the largest providers of skills training relevant to data and AI talent pipelines. EdTechs privately run short courses, modules, and bootcamps in data science, AI engineering, and coding. The acceptance of certifications over formal degrees differs by industry and employer, but the rise of these private and agile training modalities is bridging the capacities of traditional institutions and the rising need for data and AI professionals in the ecosystem. T4SU program supports access to high-demand digital training that is aligned with startup needs by partnering with EdTechs such as GoMyCode.



ESO tip | Bridging the "research-to-market" gap

Universities often operate in siloes. ESOs should coordinate with universities and act as intermediaries, helping startups access university hardware and historical public datasets to prototype products before they hit the market.

Chapter 3. Data and regulatory environment

The long-term viability of data and AI-driven agriculture depends on the foundational policies and digital public goods. This chapter explores the availability of open-access datasets and supportive regulatory frameworks that dictate how easily data and AI solutions can be developed and deployed within a particular national context.

3.1. Public data for the agriculture sector across the eight study countries

Public agriculture datasets serve as the foundational enablers for training, validating, and scaling data and AI-driven innovations. The availability of agricultural data, such as satellite imagery, market information, and soil health data, can lower entry barriers for AgTechs and develop better and more competitive solutions. The map visualises public data portals and platforms with agriculture datasets capturing different types of data, such as agriculture production, market prices, and climate/weather.

Figure 13: Public agriculture datasets map

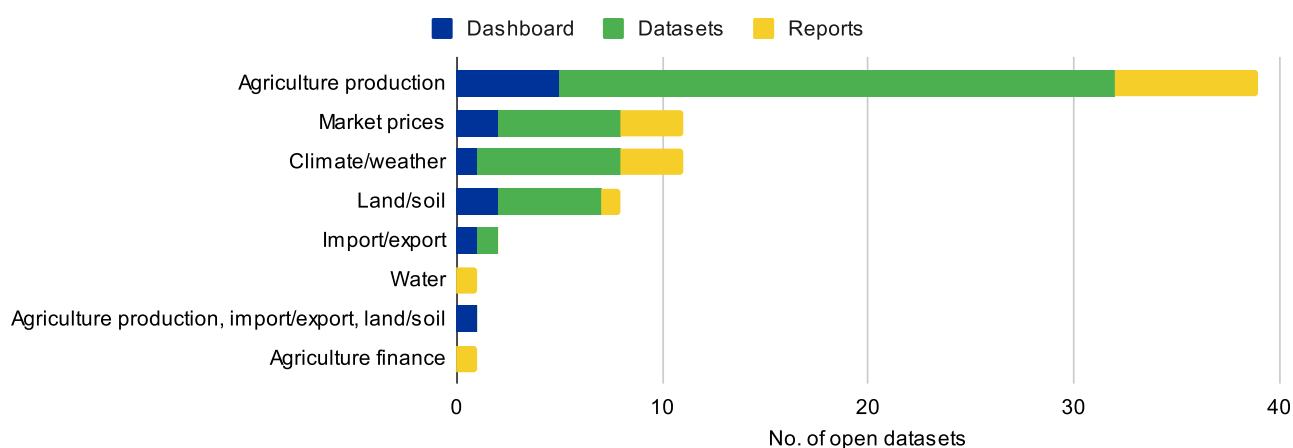


Source: Briter Intelligence and AgBase (2026)

The mapping for this study identified over 70 datasets on agricultural activities that are open for public access across the eight markets. The majority of the datasets provide data on agriculture production, followed by climate and weather data, market price information, and land or soil information. Few datasets report on water availability, import and export data, and agriculture

finance information. Most of the countries in the study share either raw data on agriculture production or some aggregate figures through reports and dashboards. The majority of institutions hosting or updating the agriculture open datasets are government ministries and publicly funded research institutions. Examples of these datasets are annual agriculture sector indicators by the Agence Nationale de la Statistique Côte d'Ivoire⁶ and IMPACT Lite data on Ghana by International Livestock Research Institute (ILRI)⁷.

Figure 14: Number of open datasets by topic and access method



Source: Briter Intelligence and AgBase (2026)

Three consistent observations emerge from the public agricultural records. Firstly, information is frequently outdated, as many datasets rely on one-off projects rather than regular monitoring, leaving a significant gap between data collection and its eventual release. For example, the annual agriculture sector indicators by the Agence Nationale de la Statistique Côte d'Ivoire jump from 2021 to 2022 data to 2025, missing years of data in between. Secondly, there is a lack of local detail, with records often restricted to broad national averages that fail to capture the specific soil or crop variations needed for precise farm-level tools. Finally, conflicting standards between government departments create a significant hurdle; without a shared naming system or common "ID" for farmers and land, startups must invest substantial time and money simply to make data from different ministries work together. **These three observations show that the eight countries have public institutions sharing information, but they are not yet sharing usable, open data.**



Policy tip: Incentivising open or common data

To prevent data siloes, governments should incentivise startups to share anonymised or aggregated insights with the public sector. In exchange, the state can offer some sort of credits or access to high-value public infrastructure (like national data centres), ensuring that private innovation also fuels public good.

6. [Annual agriculture sector indicators, Agence Nationale de la Statistique Côte d'Ivoire](#)

7. [IMPACT Lite data on Ghana, International Livestock Research Institute \(ILRI\)](#)

3.2. Policy and regulatory actors across the eight study markets

Public institutions influence the data and AI-driven innovation landscape by setting the broader direction of innovation priorities, setting standards, and enforcing regulations. The map below shows policy and regulatory actors that are developing, advising, and enforcing data-related policies across the eight focus countries.

Across the regulatory bodies, several key policies are emerging to shape the data and AI-driven AgTech innovation landscape. There are over 30 policy and regulatory actors identified across the eight countries whose role is to develop and implement regulations, provide oversight on policy enforcement, and set industry standardisation. Most of the policy regulations around data and AI happen at the ICT ministerial level. All eight countries also have ministry-affiliated or independent bodies that oversee policy implementation, and at least one agency that specifically oversees data protection laws. The section below outlines some of the key policy areas currently influencing the enabling environment for data and AI-driven startups. Relevant country-specific policies for each of these areas can be found in the [Annexe](#).

Cross-border data sharing and regional open data policy development are underway through the African Union Data Policy Framework (AUDPF)⁸. The ongoing framework draft aims to harmonise data policies across its member states to enable more regional data sharing. The framework focuses on three key components: (1) a data categorisation and sharing framework, which establishes standards to protect sensitive and critical data while enabling publicly accessible data, (2) a cross-border data flow framework, which focuses on the technical enablement and facilitating collaboration to share data across the member states, and lastly, (3) the introductory open data strategy across member states to encourage more transparent public data systems. These frameworks are guiding principles for each of the states and serve as the starting point to achieve a Digital Single Market (DSM).

Building a use case on cross-border data practice

The Ivorian Ministry of Forestry enabled access to satellite images for the AI4SU Hackathon. Tolbi, a Senegalese data startup that has built satellite-based forestry monitoring to capture biomass, deforestation rates, and carbon stocks in real-time, participated in the AI4SU Hackathon and tested their model using Ivorian datasets. Tolbi was able to correlate satellite data on Ivorian cocoa forestry to estimate carbon sequestration for agroforestry projects that were previously invisible to carbon buyers. This example showcases that enabling cross-border data through private-public partnerships can enable a practical use case in the AgTech sector.

Data privacy protection laws are integral to equitable and inclusive data and AI tech development and adoption, as they set the standards and enforcement mechanisms to protect the vast amount of personal data that is often used to train AI models. Data privacy laws are designed to bring more

8. [African Union Data Policy Framework \(AUDPF\)](#)

transparency and clarity on how businesses should use, store, and share their data for the model. All eight countries show data protection bills, but differ in their approach. Ghana and Kenya focus on the right to explanation and the mechanisms for the human appeal process, while Tanzania, Rwanda, and Senegal have taken on a data localisation approach, and Côte d'Ivoire and South Africa have taken on a more process-centric approach that mandates DPO registration and impact assessments.

Data governance enforcements have strengthened across African countries in recent years. One area of improvement is in clarifying the data governance parameters and processing expectations. Examples include South Africa introducing a stricter 30-day processing timeline for addressing claims to personal information, and Kenya introducing a data sharing code framework for lawful data sharing between entities and guidance on dealing with sensitive information, including children's data and health data. Regulators across the eight countries also demonstrate more enforcement measures. Kenya's Office of the Data Protection Commissioner (ODPC) published 30% more decisions in 2025 compared to the previous year⁹, including issuing a fine to companies that did not comply with data protection laws. Ghana also issued a deadline for data controllers to comply with data privacy laws.

Data localisation and cloud sovereignty policies stipulate that locally generated data be physically stored in servers within the borders of the country. The rationale and expected benefits of these policies include applying local data protection law to entities, incentives to build domestic cloud and server capacity, and lower latency. But data localisation regulations are contentious, as cloud localisation can increase the cost of data storage for domestic startups. The price of cloud solutions can be higher in some countries, and cross-border scaling companies would need to find cloud solutions in every country they enter. These localisation policies need to be coupled with strategic investments into local data centres, stabilising or improving energy sourcing to data centres, and improving overall connectivity to reap the benefits of these policies. Currently, Rwanda and Tanzania require localisation of data, with Senegal and Ghana likely following suit. Kenya, Côte d'Ivoire, and South Africa are testing data sovereignty and localisation laws on sensitive data such as health and personal data.



Startup note | Strategic data handling

In countries with strict storage laws (like Tanzania), keeping all data on local servers is expensive. To save costs, process as much data as possible on the farmer's device and only send the most essential, anonymised data to the cloud.

Data interoperability standards facilitate better data sharing across data controllers by introducing taxonomy, technical format or collection timeframe standards. In Kenya and South Africa, data interoperability often shows up through setting clear data sharing codes and penalising data controllers that do not create data interoperability. Rwanda uniquely leverages its digital ID infrastructure by mandating that AgTechs match the farmer data to the digital ID registered in the

9. The [ODPC's 2025 determinations](#) numbered over 150, compared to 118 in 2024.

Ministry of Agriculture. Senegal and Ghana are in the earlier stages of developing interoperability standards, and they are starting with specific market segments such as voice-first metadata standards.



Policy tip: Standardising the "language" of data

Data interoperability relies on a common language and vocabulary. Adopting global standards like AgroVoc ensures that "Maize" in one system isn't interpreted as a different variety in another, facilitating seamless data exchange across the value chain.

AI strategies have been emerging across the focus countries to set longer-term governance strategies to develop better AI technologies and talent, to transition existing industries to an AI-ready future, and to strengthen the public sector capacities for this transition. Across the eight countries, many of the AI strategies are related to infrastructure development for more sustainable energy security, ensuring better internet connectivity, and attracting more investment to build additional data processing capacities. South Africa and Rwanda set infrastructure development at the centre of their AI strategy, while Ghana and Tanzania prioritise ethical innovation that will close the digital AI gap for the broader population. Kenya and Senegal aim to create a more competitive commercial landscape and develop a regional hub. Many of these policies are new or recent, meaning the strategies have not yet fully translated into policy or investment actions, but indicating strong intentions to build more proactive approaches to embed AI in national agendas.



Policy note | Sequencing for sustainability

Avoid leapfrogging the necessary steps needed to create a data-driven environment. A "digital-first, data-second" approach ensures that connectivity and value chain digitisation are robust before introducing complex data or AI-specific governance.

IP protection laws safeguard copyright and licensing for new AI and data science advancements. IP protection at an industry level and at research institutions can incentivise innovation and information sharing. Many of the IP protection laws have been introduced in the last year, following the African Union's Continental AI Strategy in 2024, which directs member states to update their national IP laws. This means that most countries reviewed their IP practices, but the exact implementation of these laws is still being formulated. South Africa has a longer IP regulatory history, and the IP regulation update focuses more on clarifying the legal boundaries between human and machine creators. In Tanzania, Botswana, and Kenya, the IP regulations take on a more defensive approach designed to protect local data and sensitive information from being harvested against local regulatory protection. Rwanda, Senegal, and Ghana take on a more offensive approach that prioritises data sovereignty enforcement and introduces a safer environment for AI technologies to develop.



Startup tip | Data as a strategic asset

In ecosystems with limited open data, unique datasets on soil health and land form part of a company's intellectual property. While contributing to open data standards benefits the sector as a whole, it is also important to understand how you protect the raw, high-granularity data that forms your competitive advantage.

Data and AI regulation sandboxes provide safe but supervised spaces for data and AI-driven startups to test their products, exempting them from existing regulations that may stall their innovation. While regulatory sandboxes for AI are still very nascent in Africa and their efficacy is too early to observe, Rwanda has been leading the test of a regulatory sandbox through its dedicated AI Sandbox under the National AI Policy (2023). The Rwandan AI sandbox focuses on AgTechs and HealthTechs to develop new technologies with temporary exemptions from data localisation requirements by using the National Data Centre. While the outcomes are still early, the sandbox has informed the formation of the 2024 IP Protection Law and has helped develop audit and certification mechanisms for AI startups. Kenya is also trying out a sandbox to monitor AI models that use Kenyan biometrics, and noticeable AgTechs such as Digital Green, Apollo Agriculture, as well as its flagship soil health virtual agronomist project, have gone through the sandbox.



Policy note | Using sandboxes for market signals

Use regulatory sandboxes to test the impact of localisation laws on startup operational costs. This allows for more agile adjustments and iterations before full-scale enforcement, preventing unintended barriers to entry.

Chapter 4: Diagnostic framework

4.1. Introduction of the diagnostic framework

To assess the readiness of each country to support a data and AI-enabled AgTech ecosystem, a multi-dimensional maturity framework is adopted to benchmark how countries are moving from nascency to maturity. The diagnostic framework leverages the data analysed in the previous sections as well as additional macroeconomic data to assess the technical, legal, relational, and financial aspects that influence the data and AI-driven AgTech landscape. A combination of data leveraged from Briter Intelligence on the AgTech innovation ecosystem, macro indicators, and more qualitative insights from desk research and stakeholder engagements informs the data inputs into the diagnostic framework.

This section analyses the eight focus countries included in the study: Botswana, Côte d'Ivoire, Ghana, Kenya, Rwanda, Senegal, South Africa, and Tanzania. In addition, Namibia's macroeconomic conditions are analysed to create a comparison point for Botswana and other nascent markets, but it should be noted that Namibia is not included in the other assessment categories¹⁰.

The diagnostic assessment categories used in this framework are:

● **Ecosystem development:**

This category assesses the state of the AgTech innovation ecosystem development by sizing the AgTech market, technical-readiness, the support mechanisms existing in the ecosystem, and the financing into the ecosystem.

● **Macro environment:**

This category assesses the relational economic and agricultural conditions that serve as the baseline for the market conditions and agricultural sector in which data and AI-driven AgTechs operate.

● **Enabling environment:**

This category measures regulatory, legal, and data infrastructure that enables data and AI technology development across the broader innovation ecosystem, as well as more specifically in the agriculture sector.

10. The assessment across eight countries diagnosed Botswana as the only nascent ecosystem. To provide a richer perspective of what constitutes a nascent ecosystem an additional country was introduced. Namibia was selected as a benchmark country with similar characteristics to Botswana due to its resource concentration to capital-intensive mining, public sector's interest in diversifying its industries, demographic sparsity, and its geographic proximity to South Africa, meaning it also experiences some "spillovers", such as talent, domain knowledge, and introduction of technology. In addition, Namibia is also at the earlier development of open data and data regulations. These factors make Namibia a good comparison to Botswana in the diagnostic. Because this was a targeted exercise to strengthen the Botswana diagnostic from a macro perspective, further investigation into Namibia across ecosystem and stakeholder data was not pursued.

Each country is given a score based on its performance against indicators under each of the dimensions above. An in-depth breakdown of the different indicators and the scoring mechanism used to allocate the countries into different archetypes can be found in the [Annexe](#). Based on the combined scoring from the diagnostic of each dimension, the eight countries are assigned to one of **four archetypes**:

- **Nascent ecosystems** are currently at the primary stage of digital development, where the objective is to build the essential bedrock of a technology economy. These markets are typically characterised by low levels of overall innovation activity and significant connectivity gaps, particularly in rural agricultural zones. Innovation at this stage is largely state-led or activated by international donor initiatives, and private sector involvement in the funding and development of AgTech remains limited. Ecosystem support mechanisms tend to be generalist in nature, focusing on basic business administration rather than deep-tech or AI-specific mentorship. For countries in this category, the focus is on creating digital public goods such as farmer registries and connectivity infrastructure that are required before more advanced data and AI-driven solutions can take root.
- **Activation ecosystems** are larger than nascent ecosystems but are still early in their progress towards more established innovation and funding landscapes. In these ecosystems, governments typically adopt a more proactive, hands-on approach to activating the ecosystem at large, and the ratio of publicly funded or initiated programmes is proportionately larger than the size of the ecosystem.
- **Market-responsive ecosystems** exhibit growing signs of a maturing innovation landscape, with a proportionately larger volume of early-stage startups and a burgeoning group of growth-stage AgTech companies. These markets are distinguished by the rise of specialised ESOs that provide technical mentorship specifically for data and AI-driven agriculture. These markets also utilise more targeted governance or regulatory approaches that respond to the existing market gaps and needs.
- **Scaling ecosystems** possess considerably more developed and dense innovation landscapes, featuring a diverse range of companies moving from early-stage development toward market exits or maturity. These markets are characterised by a high degree of technological specialisation, where AgTech solutions are often deeply integrated with broader DeepTech capabilities like satellite-AI fusion and agentic automation. Private sector engagement is robust, with venture capital and corporate R&D playing a leading role in funding and representation, while the public sector shifts its focus to more complex regulatory duties, such as AI ethics and data sovereignty. In these ecosystems, the support infrastructure goes beyond fostering an innovation ecosystem, but focuses on larger-scale infrastructure development that can bring in high-performance computing and energy resilience required to sustain large-scale data and AI operations.

In addition to the three archetypes identified above, there is an additional mature ecosystem archetype where agriculture data is mostly open and accessible through a centralised system, data-driven agriculture activities are highly digitised, and the regulatory environment highly encourages data sharing while protecting data privacy. The countries in the diagnostics have not yet been identified as the mature archetype.

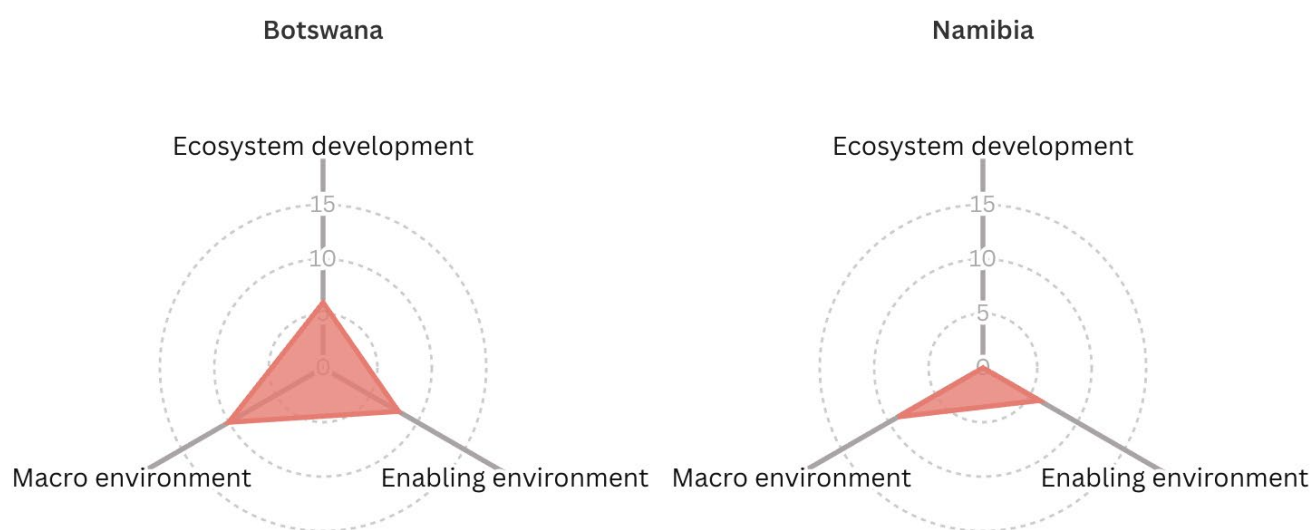
Based on the data collected, the countries are categorised on a grading scale from nascent to scaling according to the scoring framework (see [Annexe](#)). Note that within each maturity archetype (nascent, foundational emerging, Market-responsive emerging, scaling).

Figure 15. Diagnostic framework and scoring

	Botswana	Namibia	Côte d'Ivoire	Senegal	Tanzania	Ghana	Rwanda	Kenya	South Africa
Ecosystem development	6	N/A	10	18	16	16	18	18	14
Macro environment	10	9	14	12	14	13	11	14	14
Enabling environment	8	9	10	11	10	10	10	14	16
Combined archetype	Nascent		Activation			Market-responsive		Scaling	

4.2. Nascent ecosystems: Botswana and Namibia

Figure 16: Nascent ecosystem diagnostics



Source: Analyst calculations (see [Annexe](#))

Note: Namibia's ecosystem readiness and enabling environment score are low due to limited data.

Overview

Botswana and Namibia are characterised by low levels of overall innovation activity and digital connectivity gaps in rural zones. Innovation is largely state-led or activated by international donor initiatives. Ecosystem support tends to be generalist, focusing on basic business development across the AgTech sector. Both countries need to take a **“digital first, data second”** approach to activate the foundational ecosystem.

Botswana is a small market with relatively high per-capita wealth and strong government funding into public education, but its AgTech scene remains nascent due to a historical reliance on the mining sector. The Botswana Government's ambition is to diversify from a resource-intensive economy to a more diversified one. To support this effort, the government-funded University of Botswana is starting to evolve from teaching and learning to research and skills development. However, Kim Cornfield, the Director of R&D at the University of Botswana, signals that this transition is still relatively new.

Compared to Botswana, **Namibia** also has a similar economic background of high reliance on mining, proximity to the South African markets, small population, and low levels of innovation ecosystem maturity. However, Botswana showcases a higher level of digital connectivity in rural areas compared to Namibia¹¹ and the government's regulatory landscape focuses on IP and sovereign data access compared to Namibia's focus on climate resilience.

11. [UNESCO Global AI Ethics and Governance Observatory. Botswana country landscape.](#)

● Ecosystem development

Ecosystem development is focused on building technical readiness. A nascent innovation ecosystem is characterised by state-led initiatives and early-stage infrastructure. A digital-first approach requires moving from legacy systems to a modern technical floor. As noted by Kimberly Cornfield, layering data and AI is premature when the hardware at the University of Botswana is mainly from the pre-2010s and needs to be updated. Foundational digital tools like electronic signatures or automated workflows are not yet utilised in the ecosystem.

● Macro environment

Botswana is focusing on **developing the infrastructure for increasing digital literacy and digital accessibility across its agricultural and rural communities.** Botswana's mobile subscription has been increasing rapidly over the past few years, and its SmartBots initiative connects mobile users to more secure servers. Namibia, on the other hand, has lower mobile subscription numbers than Botswana, and the country's strategy is more focused on expanding internet coverage across rural zones.

● Enabling environment

Botswana has a few publicly published datasets on agricultural activities. Most of them are aggregated datasets on agriculture production, made available across several public institutions, including the national statistics agency Statistics Botswana, Botswana University, and agriculture research institutions. However, many of the datasets are not updated regularly, and the data is shared in multi-year aggregates, making it difficult for AgTechs to access up-to-date data in an accessible format.

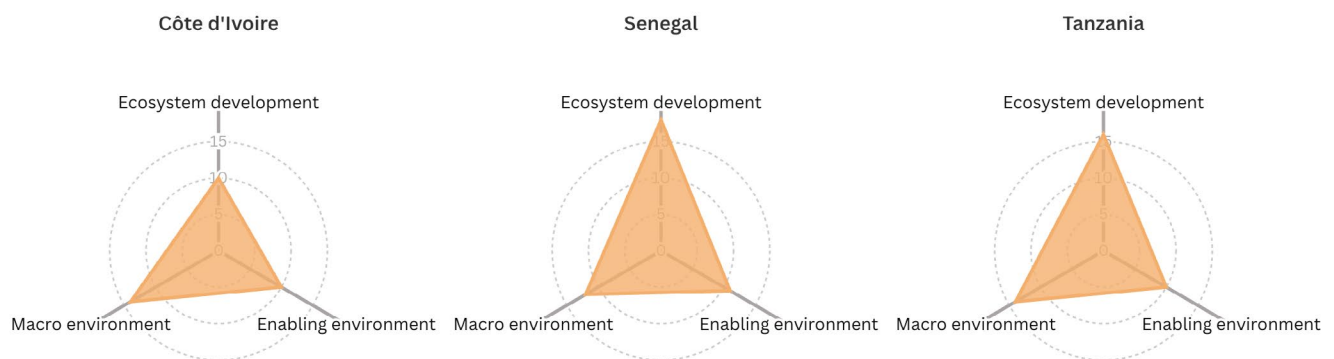
While it can be premature for nascent ecosystems to develop AI-specific policies, **the presence of an AI strategy and baseline policies on IP and data protection law signals the direction of AI technology development.** Botswana has integrated AI strategy into the SmartBots initiative, which focuses on increasing rural connectivity, and the National Transformation Strategy, which digitises government functions across institutions.

4.3. Activation ecosystems: Côte d'Ivoire, Senegal, and Tanzania

Overview

Activation ecosystems that have grown their innovation ecosystem and digital infrastructure beyond the nascent stage, but are still in earlier stages of ecosystem growth. These ecosystems show efforts to develop the foundational infrastructure for stimulating and growing innovation ecosystems, such as talent training and ESO network development. The key characteristic is a smaller and earlier stage of innovations in the data- and AI-driven sectors, which means the support ratio per company is higher. These countries develop agriculture open data to test specific use cases that can feed into the broader growth of their policy support for growing the ecosystem. **Senegal, Tanzania, and Côte d'Ivoire** are within this category.

Figure 17: Foundational emerging ecosystem diagnostics



Source: Analyst calculations (see [Annexe](#))

Macro environment

Agriculture is a large sector across activation economies that employs a large number of the population. Across the 3 countries, agriculture, forestry and fisheries make up 20% of the overall GDP value, and employ over a third of the population.

However, the internet access is still mainly through mobile phones, and secure internet broadband access remains limited. Most activation archetype markets have fewer than 70 secure servers per million people. But a high level of mobile subscription shows these are predominantly mobile-first ecosystems.

Across the emerging markets, the high cost of data and AI development is hindering growth for AI adoption and development. The AgTech sector is already a difficult sector to be revenue-driven due to the income constraints of farmers and fragmented market access. On top of that, Daniel Ayenoue of Orange-Sonatel Startup Lab points out that "real-time, API-downloadable data at a national level is non-existent, meaning that AgTechs spend a lot of time and resources to build their own datasets from scratch. While this creates a competitive advantage for those who succeed, it raises the cost of data". It also increases the entry bar for newer, smaller innovators. Daniel further remarks that many projects fail post-pilot because the cost of maintaining clean, high-quality data pipelines becomes prohibitively expensive.

Ecosystem development

Senegal, Tanzania, and Côte d'Ivoire are smaller, and the funding for these ecosystems remains limited. Across these three markets, there are fewer than 30 AgTechs identified across these emerging markets, and visible funding remains very limited. Particularly, funding from commercial funders, with the exception of Tolbi from Senegal, remains less visible. This means the policies and enabling efforts are focused on building and developing the foundational size and infrastructure for the innovation ecosystem at large, and they may not be as market-driven as there are not yet a sizeable number of market players to drive industry voices.

Enabling environment

Senegal, Tanzania, and Côte d'Ivoire are focusing on activating the market, such as improving talent development, and creating more data on agricultural activity.

Three markets have a higher proportion of ESOs and open agriculture datasets compared to their small market size. Most of the datasets are on agriculture production data, such as the Côte D'Ivoire Data Portal's Agriculture Sheet¹² and Tanzania's Agri-Inputs Platform¹³. These datasets are open for public use and are accessible mainly through aggregate figures on reports, and some have downloadable data.

However, **the above agricultural datasets are not updated or consistent across the years.**

For example, the Côte D'Ivoire Data Portal's Agriculture Sheet covers 2008 to 2013, but does not host data past 2013. Similarly, Tanzania's Agri Input Platform's directories are still at the earlier stages of development. The emerging ecosystems need to build more trust and centralisation by providing up-to-date information that follows global data formats and sharing structures. For example, mature ecosystem players such as the Korean Institute of Agricultural Sciences make comprehensive soil health data available¹⁴ including detailed soil maps, data taxonomy, and GIS maps across Korea on an annual basis.

For the markets that are building the foundational capabilities, talent retention and brain drain remain a challenge.

Ismaila Seck of AI Hub Senegal notes the primary challenge is "securing financial resources to attract and retain quality talent," as engineers are frequently lured away by hubs in Kenya, South Africa, or European markets. AI Hub Senegal facilitates a programme where it subsidises 50% of intern fees for six months to help local startups match the financial lure of larger companies.



ESO note: Combatting "brain drain"

To retain high-tier engineering and data talent, ESOs should move beyond generalist business mentoring to provide more targeted training. Focus on creating science-to-business pipelines that support technical R&D and access to resources for companies taking a data-driven approach.

12. [Côte D'Ivoire Data Portal's Agriculture Sheet](#)

13. [Tanzania's Agri-Inputs Platform](#)

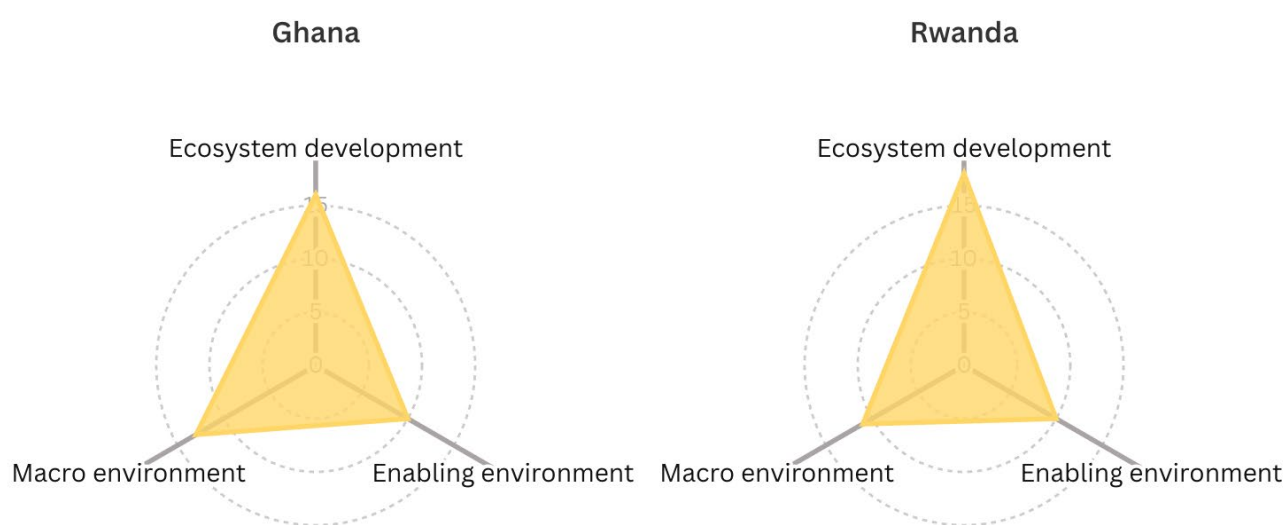
14. [Korean Institute of Agricultural Sciences soil health data](#)

4.4. Market-responsive ecosystems: Ghana and Rwanda

Overview

Market-responsive ecosystems have relatively larger innovation ecosystems, with some AgTechs gaining some traction from private and public sector funders. These ecosystems leverage their market activities and existing digital infrastructure, such as digital national ID, to design broader data interoperability policies and data initiatives. Ghana and Rwanda are categorised as market-responsive emerging ecosystems whose policy and enabling environment is responsive to growing and expanding the existing innovation ecosystem, rather than focused on the foundational market creation compared to foundational emerging ecosystems.

Figure 18: Market-responsive emerging ecosystem diagnostics



Source: Analyst calculations (see [Annexe](#))

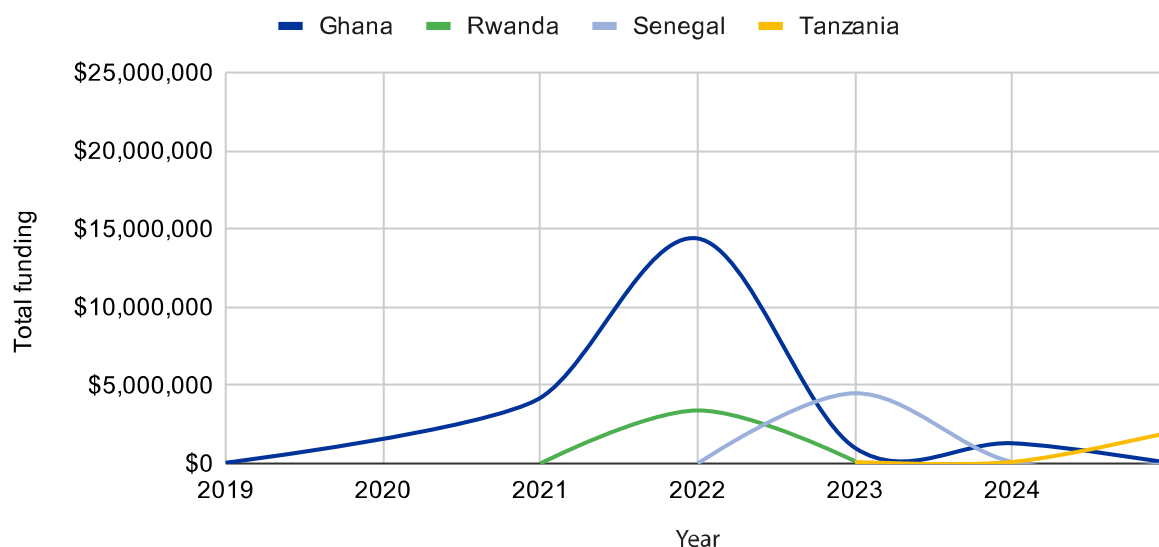
Macro environment

Internet access for mobile and non-mobile users is growing across the markets. In particular, Rwanda's secure internet access is much higher than other African markets at 160 per 1 million people. These numbers are still much below larger markets such as Kenya, at 400 per 1 million, or South Africa, 16000+ per 1 million people.

Ecosystem development

Ghana and Rwanda have a larger market for data and AI-driven innovation. Ghanaian data and AI-driven AgTechs have been emerging earlier than other ecosystems and have demonstrated traction earlier than other markets. One example is Farmerline, which connects smallholder farmers with quality input and market information by linking input providers, brokers, and farmer data on its Mergdata platform. Farmerline has developed corporate partnerships and has scaled by providing a good use case solution to farmers across West Africa. Rwanda follows after Ghana and saw more commercialised data-driven AgTechs emerge starting in 2021. Both Ghana and Rwanda have a higher proportion of MSMEs compared to other emerging markets, which indicates a more entrepreneur-friendly environment.

Figure 19: Data and AI-driven AgTech funding trend across focus countries



Source: Briter Intelligence and AgBase (2026)

Enabling environment

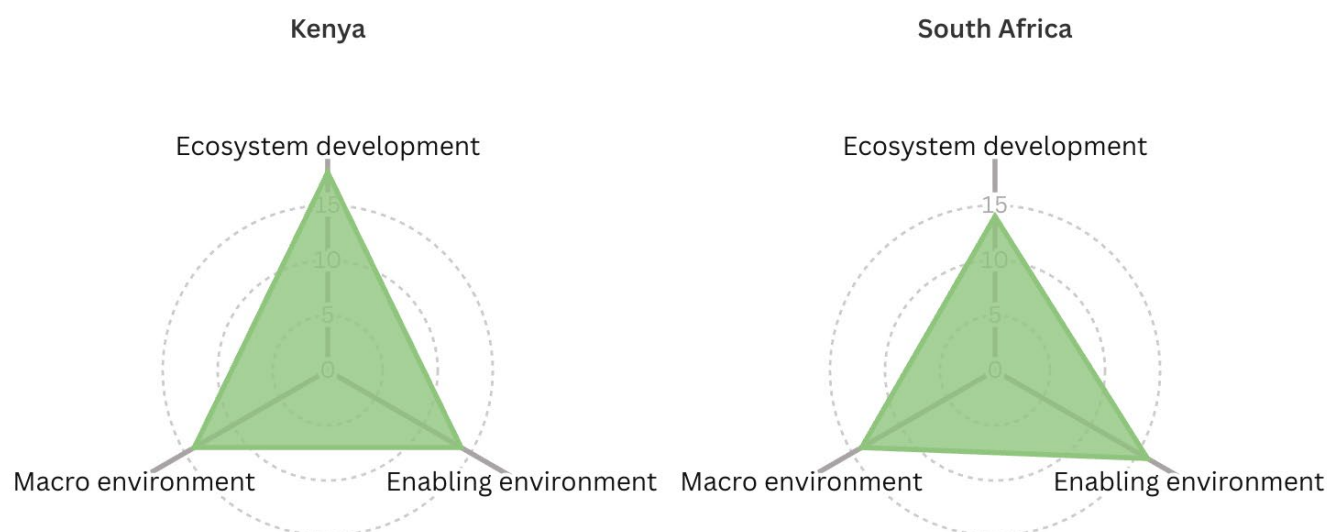
Ghana and Rwanda are leveraging the existing market players to enable specific growth areas within the ecosystem, such as improving data interoperability and data localisation.

Ghana has introduced an inter-governmental IT interoperability framework in 2022, developing a foundational national identity card that functions as a central identification mechanism to access other government and functional services such as health service, travel, and administrative tasks¹⁵. Rwanda has taken a similar e-identity approach with a centralised digital identity card, centralised digital financial service facilitation, and building sectoral services such as health or farmer registries that require intensive inter-agency data sharing capabilities. Rwanda has also introduced strong data localisation rules, but has leveraged a regulatory black box for data and AI-driven companies to test the feasibility of the localisation regulations.

14. <https://www.hsrbg.org/wp-content/uploads/2024/09/Integration-and-Interoperability-of-National-Identification-Systems.pdf>

4.5. Scaling ecosystems: Kenya and South Africa

Figure 20: Scaling ecosystem diagnostics



Source: Analyst calculations (see [Annexe](#))

Overview

Kenya and South Africa, two of the key digital hubs on the continent, represent the scaling ecosystem archetype for data and AI-driven AgTech development. Kenya and South Africa have a comparatively larger, more complex, and connected AgTech ecosystem funded by a diverse set of actors ranging from private, concessional, and public funding. They function as regional hubs with access to high-quality data and AI-relevant talent, expertise, and localised capital sources. South Africa stands out as a technical hub with its presence of advanced corporate research, university-led AI and data research, and high-performance computing clusters. Kenya is more of an agriculture sector hub with a cluster of national and international agriculture research and a robust AgTech ecosystem. In both countries, the governmental focus is shifting to strengthening the infrastructure for increasing demand for AI, safeguarding IP and privacy law, and attracting further private sector engagement and investment into the sector. These scaling ecosystems are where specialisation on the cross-section of AgTech and AI can emerge, thanks to the concentration of relevant actors in both areas.

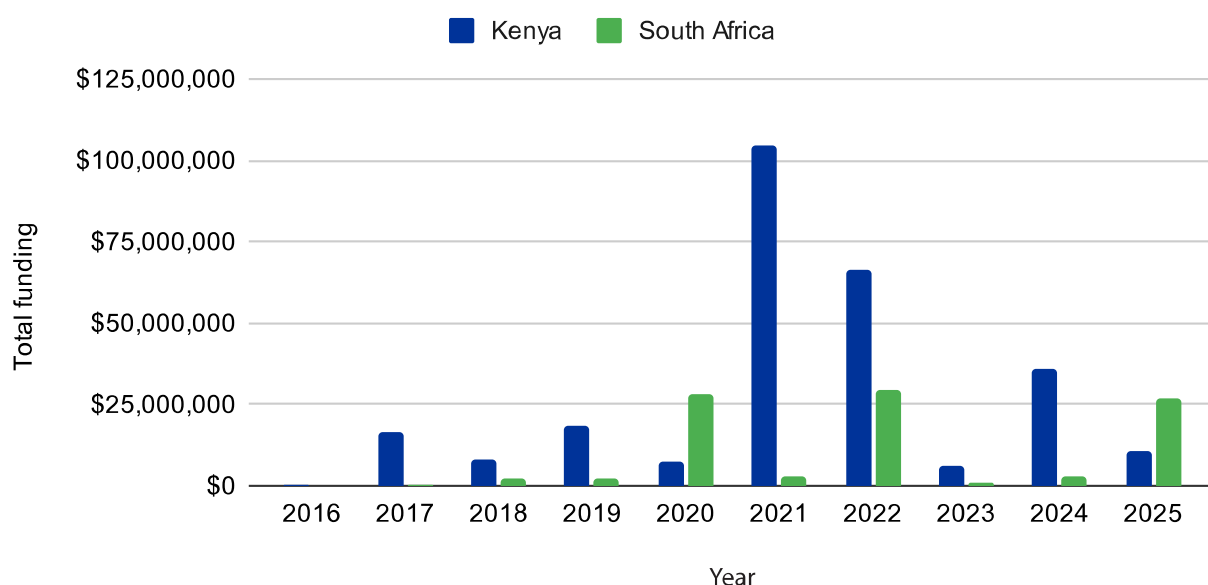
Ecosystem development

Data and AI-driven AgTechs across African scaling ecosystems have raised over \$370 million¹⁶ over the last decade. Even though the majority of this funding is by a few scaling AgTechs that fundraised above \$10 million, these scaling companies are demonstrating the traction and commercial viability of data and AI-driven AgTech companies. In these scaling ecosystems, the funder landscape also diversifies from being mainly public sector-driven to a mix of concessional and commercial funders. While there is still a big presence of concessional funders that support early activation efforts through ESOs, competitions,

16. One third of disclosed funding records do not have disclosed funding size. So the total size of funds raised in these markets are likely to be larger than what has been recorded here.

and the grant opportunities, venture capital and SME funds are more active starting at the seed stage. Kenyan later-stage deals show more blended financing instruments with a mixture of foundations, DFIs, and commercial investors, while South Africa shows a little more corporate participation along with VCs and foundations.

Figure 21: Funding trend across scaling ecosystems



Source: Briter Intelligence and AgBase (2026)

Increased data availability and AI talent enable AgTechs in the scaling markets to improve efficiencies across their business operations and create better services for the end users. Innovators are moving beyond basic notification systems to develop autonomous agents capable of independent operational tasks. Rhea Africa, an AI-powered agronomic advisory service for Kenyan farmers, successfully built the AI capacities to transition from manual agronomy reporting to generative AI models. Rhea reduced reporting times from hours to near-instantaneous delivery, while expanding into computer vision for crop disease analysis. Recently, Rhea also built an AI-powered chatbot capacity to respond to increasing customer service demands along with VCs and foundations.

Macro environment

AgTech is a much more strategic and important sector in Kenya compared to South Africa. The macro-relevance of AgTech is far higher in Kenya, where the sector contributes over 20% of GDP and employs a third of the population. In comparison, agriculture is less than 5% of the South African GDP. This implies that the development and investment in AgTechs in Kenya is a strategic necessity, while South African AgTechs are a part of the broader innovation ecosystem development and optimising different segments' value chains.

In Kenya, the key constraint is developing a secure server infrastructure to support its growing data-driven ecosystem. Kenyan farmers and the rural population are still mainly

connected via mobile devices and have a relatively good connection to mobile financial services, but data and AI-driven AgTechs are likely still relying on foreign servers instead of domestic servers, this will likely impact the long-term cost for AgTechs as well as the data sovereignty of farmer data. South Africa has 40 times more secure servers per million people compared to Kenya, with some additional data centre development underway.

● Enabling environment

Despite these increased open data efforts, the cost of reliable data is still high in scaling markets. Even in Kenya, where there are considerably more agriculture-related open data sources than in other markets in the study, open data on specific crop types and environmental data are often fragmented and limited. This makes it time-consuming and costly to integrate existing information. Environmental data is often neither public nor accessible, leading multiple institutions to duplicate collection efforts and resulting in data gaps and poor quality that requires lengthy cleaning. The lack of data-sharing incentives is primarily due to the high cost of data collection, which AgTechs view as a crucial asset they are unwilling to share.

Coordination across government ministries focuses on making data protection and data sharing consistent. In Kenya, the ODPC is introducing sector-specific guidance for agriculture and coordinating with agriculture data agencies such as KALRO to reduce regulatory blockages relevant for AgTechs. South Africa is introducing a new coordination body through a presidential commission of 4IR to standardise the regulatory support and R&D tax incentives for startups across the country.

Clear data protection standards are needed to incentivise data sharing at the industry level. Currently, many AgTechs and private data developers are disincentivised to share their data because of existing data scarcity and data being an important asset for their unique value proposition. But in addition to that, the lack of clear data protection policies and enforcement also hinders data sharing even when there are clear opportunities for collaboration. Kenya has developed an ODPC data sharing code to encourage research and commercial collaboration, and South Africa has introduced clearer guidelines for data protection and a centralised eServices Portal to report any breaches in 2025. In both cases, the implementation and the impact need to be closely monitored to assess their effectiveness.

Chapter 5: Recommendations

Based on the ecosystem mapping and diagnostic framework, several strategic recommendations emerge. This chapter proposes a practical way forward for policymakers, ecosystem supporters, and AgTechs. The chapter first introduces cross-cutting recommendations across countries, followed by country-specific recommendations tailored to each country's stage of ecosystem maturity.

Cross-cutting recommendations across all markets



Policymakers

- **Sequencing the data capabilities.** A data and AI-driven ecosystem is built with data as the foundational block. But this requires incremental changes that reflect the local conditions and market maturity level. The sequence builds from the digital-first phase to improving data collection, standardisation, the overall quality of the data, and then making data more accessible. Based on these improved data capabilities, applied technology capabilities on AI are built. Policymakers should partner with the private sector to build and test use cases and develop capabilities across these sequential steps.
- **Talent retention and industry pipeline.** There are over 100 universities or training institutions for data and AI-related disciplines across the eight African markets studied. There is growing talent across the continent, but they need to be better connected with the innovation ecosystem to avoid talent drain. Some practical policy areas are (1) developing data stewardship capabilities within universities, (2) building working groups with developer communities to assess incentive gaps for talent to stay within the market, and (3) working with ESOs and universities to create subsidised student matching in data companies.
- **Strengthening IP and data protection.** African AgTechs often build their own data on soil health, weather patterns, and land use in order to formulate better products and resilient business models. This also means that proprietary data is considered an asset for both AgTechs and investors, and can discourage data sharing across the ecosystem. Robust data protection and IP laws are essential to provide legal security for innovators to securely share more information and attract investment. South Africa has developed a centralised IP and trademark registration process and stronger personal data security through POPIA, and the learnings from these legislations need to be shared with other countries.
- **Setting a policy framework to signal the industry.** A clear and non-restrictive national AI strategy should signal the government's commitment to AI as a key economic factor, providing the industry, innovators, and investors with a clear point of reference and market

alignment. This is preferred over highly prescriptive AI-specific policies in the early stages of market development.

Work with the industry to test policies. Most data and AI-related policies have emerged in the last few years. Many countries are getting ready to implement these policies. Policymakers need to work closely with AgTechs, investors, and ESOs to test and assess the efficacy and impact of the new policies. Governments should build policy working groups composed of investors, AgTechs, private data companies, and data infrastructure service providers to build feedback loops. Regulators should gather as much feedback from the policy sandboxes to adjust and build better policies.



Startups

Small language models (SLMs) can help build the African edge. Heavier investment into data development by AgTechs means African AgTechs are well-positioned to develop AI capabilities using SLMs. AI models developed using SLMs can develop more specialised use cases for the market and sector at a lower cost than large language models (LLMs). Continuing to build better data models can lead to more localised solutions.

Think data first, business model next, and then AI capabilities. African AgTechs likely have better competition for SLMs compared to LLMs. Data and AI-driven AgTechs need to prioritise what kind of analytics will add the best value, what kind of specialised datasets are needed, and how they can drive revenue from them rather than trying to build the company's "moat" on AI. AgTechs need to build the right business models that account for the changing price for processing and storing data. Then AgTechs need to develop a resilient data infrastructure that will comply with different data localisation rules, data ethics frameworks, and be cloud resilient.

B2B and B2G breakthroughs are crucial for AgTechs. To achieve greater scale and secure sustainable funding, AgTechs should proactively orient their business models toward B2B and B2G engagement, positioning themselves for corporate exit opportunities. There is an active market gap for national-level data on farmer advisory and data operability, and AgTechs should fill the underdeveloped public data market. Corporate acquisition and buyouts are likely exit options for many of the AgTechs. AgTechs that can establish a strong, durable data infrastructure and adopt sound cloud and DevOps practices that are suitable for the enterprise ecosystem will be better positioned for this pathway.



ESOs & funders

- **Mobilisation efforts are needed to develop more specialised support for AgTechs** by centralising expert networks, facilitating AgTech-specific knowledge sharing, and ecosystem activation efforts such as hackathons, workshops, and datathons. In addition, siloed ecosystem actors need to be brought together more frequently to establish cross-institutional collaboration.
- **Showcase success stories that emerge.** The data and AI-driven AgTech landscape is still at an early stage of development and funders need to learn more about success cases to build investor interest and confidence. Funders and ESOs should actively identify, convene, and highlight local founders and innovations that succeed. The Instadeep exit was a huge catalyst for Tunisia and is helpful in attracting investor interest. In addition, use platforms like datathons and hackathons, as done by Digital Africa and Orange Digital Center, to clearly demonstrate the tangible benefits of shared data practices and AI-driven solutions to a broader audience, including the public sector and potential partners.
- **Reframe data risk as an opportunity for business model innovation.** Investors are still learning about AI technology and the landscape. Investors should operate with the mindset that data and code are likely to become open source that necessitates business model innovation. Funders should focus on investing in companies that are building a durable and technically robust infrastructure.

Policy recommendations by market



Botswana

Ecosystem diagnostic

Nascent

No. of data and AI-startups

<10

Funding landscape

Grant-based

No. of support actors

10+

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Strength: High innovation density relative to its small size, indicating potential for high growth once activated. Challenge: The AgTech sector is in its primary stage, with very low innovation activity due to a historical reliance on mining.	Strength: High per-capita wealth and high mobile subscription rates provide a strong consumer base. Challenge: Digital connectivity gaps persist in rural agricultural zones, hindering the "last mile" of AgTech.	Strength: Strategic commitment to AI through the "SmartBots" initiative and recent updates to Data Protection (2024). Challenge: Public data is currently shared as multi-year aggregates rather than usable, real-time datasets.

Policy priorities

As a nascent ecosystem, Botswana's policy priorities should be strategically focused on establishing the foundational data infrastructure and capability. Leveraging its strong central governance, the main focus should be on strengthening digitalisation and talent development efforts to build the foundation for data development in subsequent stages.

- **Policy priority 1: Accelerate digitalisation and data collection capabilities.** utilise the strong central government's capacity to lead and invest in government-led, national-scale data collection efforts within the AgTech sector. This includes investing in state-owned sensor backbones and centralising key foundational agriculture data (e.g., soil health, weather) to ensure it is up-to-date and comprehensive.
- **Policy priority 2: Develop talent-to-industry pipeline and digital capabilities in academia.** Leverage public universities to update the technology specifications and course content to align with industry needs in data science, AI, and AgTech. In addition, universities can develop data stewardship capabilities within universities to ensure students are trained in responsible data management, quality, and governance, effectively plugging them more directly into industry jobs.

Measurable outcomes:

- Number of new or significantly updated university degree programmes/modules focused on data science, AI, or data-driven AgTech.
- Percentage increase in university graduates from relevant tech fields securing employment in local AgTech or data-related industry roles within one year.



Ecosystem diagnostic

Activation

No. of data and AI-startups

<10

Funding landscape

Predominantly grant-based early stage, with a few exceptions

No. of support actors

<10

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: A growing digital ecosystem and potential to expand into other francophone markets. Challenge: An overall small market that requires a broader digital and innovation ecosystem activation.	Opportunity: High level of agriculture sector activity and good mobile accessibility, which creates a good foundation for data-driven AgTechs. Challenge: Low secure internet server for the population, especially for the rural population, high cost of data storage, and limited spending power amongst the agricultural value chain.	Opportunity: High concentration of ESOs and support for talent development. Challenge: The data availability remains fragmented to lower the cost of data access for AgTechs

Policy priorities

Senegal is categorised as an emerging ecosystem, meaning it has made initial strides in digitalisation but needs to focus on collecting more high-quality data at a consistent timeframe to unlock the potential of its burgeoning ecosystem.

- **Policy priority 1: Expand digital infrastructure.** Continue to accelerate digitalisation efforts, particularly in rural and agricultural zones, to ensure all farmers and AgTechs have the means to collect and transmit basic data. Policy incentives, such as reducing tariffs on IoT devices, can help bring down the barrier to data collection. At the same time, the government should roll out targeted programmes to lower the cost of basic data collection tools for public institutions and invest in field agent training for quality data capture.
- **Policy priority 2: Advance data harmonisation.** National data stewards should establish common data standards and taxonomies for essential agricultural datasets used by different government and research institutions. This harmonisation effort should be coupled with creating clear incentives and secure technical mechanisms for public institutions to share data with the private sector.

Measurable outcomes:

- Number of AgTech-relevant public datasets collected or updated annually in the agriculture sector at the national level.
- Number of key agricultural data taxonomies or technical standards formally adopted by relevant government and research institutions.



Côte d'Ivoire

Ecosystem diagnostic

Activation

No. of data and AI-startups

<10

Funding landscape

Primarily grant and concessional early-stage funding

No. of support actors

<10

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: Emerging innovation ecosystem and presence of the AgTech sector at large. Challenge: Small AgTech ecosystem and a low level of data-driven innovation, and very few “Success stories”	Opportunity: High level of the agriculture sector and export industries for agricultural goods, existing baseline of mobile internet access. Challenge: Rural digital connectivity is limited, which lowers the rate of agriculture value chain workers engaging with data-driven AgTechs.	Opportunity: Focus on talent pipeline development, more ESO and enabling support for the innovation ecosystem. There are ongoing efforts for interoperability in the agriculture data governance. Challenge: Limited data accessibility and infrequent or fragmented open data on agriculture.

Policy priorities

Côte d'Ivoire is classified as an emerging ecosystem with a large agricultural ecosystem and a burgeoning innovation ecosystem. Its primary focus should be on data harmonisation and standardisation to bridge the gap between initial digitalisation efforts and a scalable data-driven economy.

- Policy priority 1: Operationalise data harmonisation and standardisation.** First, ministries should continue to establish and pilot a formal, inter-ministerial working group dedicated to data sharing and harmonisation. The primary goal is to test harmonisation efforts by selecting a few critical datasets (e.g., crop yields, real-time market prices) and aligning their taxonomies and technical standards. Second, the dedicated working group needs to enforce a set of standardised codes and quality metrics for public sector agriculture data. Incentives for public institutions to adhere to these standards will lead to the data being more reliable and usable for AgTech startups.
- Policy priority 2: Strategic private sector integration for data usability.** While Côte D'Ivoire develops data harmonisation at the ministerial level, it should launch public-private partnerships focused on solving specific, high-value challenges (e.g., pest/disease detection). These B2G opportunities allow AgTechs to test the newly harmonised data in an applied setting, which serves to both validate data quality and shape future policy.

Measurable outcomes:

- Number of core agricultural datasets successfully standardised with a common taxonomy.
- Number of active B2G pilot projects launched with private AgTech companies utilising newly harmonised public data.



Tanzania

Ecosystem diagnostic

Activation

No. of data and AI-startups

<10

Funding landscape

Primarily early stage

No. of support actors

15

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: The AgTech ecosystem as a whole is growing with diversifying products emerging. Challenge: Data-and AI-driven AgTech is a very small proportion, and not much traction has been demonstrated.	Opportunity: The agriculture sector activity is large, and the digital connectivity provides a good real-life ground for AgTech solutions. Challenge: Fragmentation and the high cost of acquiring data limit innovation in data companies.	Opportunity: High proportion of ESOs to deliver capacity building and innovation ecosystem development. Challenge: Interoperability of agricultural data remains limited.

Policy priorities

Tanzania is categorised as an emerging ecosystem with a large agricultural sector and solid levels of digitalisation. But Tanzania lacks governance mechanisms to enable better data sharing and has low gender inclusion in talent pipeline development. Improvements in these two areas can help scale their ecosystem further.

- Policy priority 1: Systematic data collection investment.** The high-value data for the agriculture sector, such as granular soil and satellite imagery, needs to be developed further through more direct investments in active, systematic data collection efforts by public institutions in high-value AgTech domains. This moves beyond passive data collection to proactive data supply.
- Ecosystem priority 2: Target capacity building aimed at gender inclusion.** The talent pipeline of women who are trained in data-related fields remains low. In order to create a more gender-balanced workforce, there needs to be targeted training and mentorship programmes for women to develop skills in data science, AI, and digital AgTech. This can include providing grants for women-led AgTech startups and women pursuing technical degrees in relevant fields.

Measurable outcomes:

- Percentage increase in the number of publicly funded, newly collected AgTech-relevant datasets released annually.
- Number of women and girls enrolled in or graduating from targeted data science, AI, or AgTech training or mentorship programmes.



Rwanda

Ecosystem diagnostic

Market-responsive

No. of data and AI-startups

<10

Funding landscape

Mainly early-stage funding

No. of support actors

20+

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: Broader tech-enabled startup ecosystem with a mixture of commercial and non-commercial funders. Challenge: Limited AgTech sector activities within the broader startup ecosystem.	Opportunity: High proportion of entrepreneurs at large, a digital government ID, including a more digitalised farmer registry. Challenge: The agriculture sector as a whole is relatively small, and localisation regulations may hinder cross-border data.	Opportunity: A deep network of ESOs for the broader ecosystem, a regulatory sandbox that feeds into policymaking. Challenge: Considering this is a small market, data infrastructure investment needs to be in parallel with the localisation efforts.

Policy priorities

While categorised as an emerging ecosystem in the report, Rwanda's advanced progress in fueling its innovation ecosystem and its specific policy requirements on data localisation create a unique challenge: managing the cost of data storage and the need for a unique competitive edge. The emphasis shifts from basic data harmonisation to cost optimisation, data localisation, and advanced AI specialisation.

- Policy priority 1: Data Infrastructure cost optimisation.** As Rwanda implements stricter data localisation policies, regulatory entities need to introduce tax incentives, subsidies, or public-private funding mechanisms to drive investment in national-level data centres. This will directly reduce the cost barrier for AgTechs, keeping them competitive while simultaneously supporting data localisation efforts.
- Policy priority 2: Develop national-level LLM specialisation.** Given the small size of the market, there needs to be a focused, government-backed programme to curate and leverage high-quality, harmonised national and localised datasets to train LLMs for the AgTech sector. This is a strategic move to overcome the small market size challenge by providing a unique, highly accurate, and valuable AI tool that gives local AgTechs a competitive edge and attracts commercial investment.
- Ecosystem priority 1: Data science talent upskilling.** Universities and training institutions should launch advanced upskilling programmes and partnerships with global AI labs and universities focused specifically on complex AI model training, data governance for large models, and the responsible deployment of LLMs.

Measurable outcomes:

- Percentage decrease in the average cost of local, certified cloud storage and processing services for registered AgTechs over one year.
- Achievement of key milestones, such as the completion of the launch of the first National AgTech LLM.
- Number of data scientists or AI developers certified through advanced training programmes in LLM development and data governance.



Ecosystem diagnostic

No. of data and AI-startups

Funding landscape

No. of support actors

Market-responsive

10+

Early-stage growth funding, a mixture of commercial and non-commercial

15+

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: Larger and more diverse startup ecosystem, including the AgTech sector Challenge: While data and AI-driven AgTechs are a small proportion, there are likely more data-driven efforts happening across the ecosystem.	Opportunity: Relatively larger entrepreneurial activities, a more secure internet server and mobile access across the country. Challenge: Still fragmented agriculture data.	Opportunity: Focused system-level challenges, such as improving interoperability rather than just building foundational capacity. Challenge: Little public-private collaboration to drive better data collection.

Policy priorities

Given Ghana's larger market size and robust innovation ecosystem support, the policy agenda should transition from establishing basic standards to refining data quality, ensuring interoperability, and maximising open accessibility. This focus positions Ghana to move from the emerging to the scaling maturity archetype.

- **Policy priority 1: Fine-tune data quality.** Ghanaian public data stewards need to move beyond basic standardisation by implementing advanced, sector-specific quality benchmarks and auditing processes for all public agricultural data. This includes introducing geographic granularity and using globally recognised technical standards across all public data repositories to guarantee interoperability. This allows AgTechs to seamlessly integrate diverse datasets without extensive cleanup.
- **Policy priority 2: Maximise data accessibility.** There should be a centralised portal dedicated to national open data that allows users to access data via APIs with clear terms of use. This directly addresses the need to make data more open. This likely needs to be doubled with regulatory and financial incentives for private AgTechs to securely and ethically contribute anonymised, aggregated datasets back into the public or shared data ecosystem.
- **Ecosystem priority 1: Industry representation to serve as policy soundboards.** Ecosystem actors need to develop a more centralised representation to provide real-time, technical feedback on new data standards and policies. These groups should also mobilise ecosystem actors to share best practices and success stories in this fast-changing ecosystem.



South Africa

Ecosystem diagnostic

Scaling

No. of data and AI-startups

25+

Funding landscape

Growth stage funding, corporate funding

No. of support actors

25+

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: A larger and mature innovation ecosystem with a wide range of ESOs, including research and universities. Specialised support for data and AI innovation is emerging Challenge: Data silo by companies to maintain unique data value within the company, as data development is still costly.	Opportunity: Secure internet access is one the highest on the continent, and shows a big uptake in the data storage and processing infrastructure. Challenge: Energy stabilisation is needed for the increasing demand for data centres.	Opportunity: Data protection laws are becoming more prevalent, giving companies better guidelines for data practices. Challenge: Data availability for agriculture data remains limited, hindering growth for data-oriented companies or promoting B2G growth pathways.

Policy priorities

South Africa is a scaling ecosystem, characterised by strong existing data and AI capabilities, robust corporate connections, and mature policy frameworks. The key challenge is connecting this advanced, corporate-focused ecosystem with the agriculture segment. The primary policy priority is enabling more public data usage for the private sector with a focus on tech specialisation in agriculture.

- **Policy priority 1: Enhance public data usage.** Enabling access to agricultural data required by the industry, such as market price data and import/export pricing data, in downloadable formats.
- **Ecosystem priority 1: Tech specialisation to cross over between data/AI and agriculture.** A dedicated public and private R&D funding for university and industry consortia focused on creating deep, specialised Ag-AI solutions for critical agricultural sector needs, such as specific crop disease modelling and data-driven water and drought management prediction.
- **Ecosystem priority 1: Highlight success cases of data-driven AgTechs.** Publicising the success cases of data and AI-driven AgTechs that have scaled to demonstrate scaling pathways and build investor confidence. The government can play a role in this communication strategy by introducing a trustworthy and centralised platform to highlight success cases.

Measurable outcomes:

- Number of public agricultural datasets made available on a centralised, machine-readable platform.
- Total value of government or public-private funding allocated to specialised Ag-AI research and development grants.
- Number of formally documented and published success cases of data and AI-driven AgTech solutions.



Kenya

Ecosystem
diagnosticNo. of data and
AI-startups

Funding landscape

No. of support
actors

Scaling

25+

Growth stage funding, DFI +
commercial funding

25+

Diagnostic summary

Ecosystem development	Macro environment	Enabling environment
Opportunity: One of the largest AgTech ecosystems on the continent, with a lot of donor and DFI interest to build use cases in the AgTech sector. Challenge: Commercialisation of data-driven and AI AgTechs is still being tested.	Opportunity: High digital penetration rates and a mature agriculture sector ripe for scaling and validating data-driven AgTech solutions. . Challenge: The AgTech sector still struggles with the cost of data collection and low willingness to pay amongst smallerholder farmers.	Opportunity: Diverse and in-depth data stewards for the agriculture sector are present in the market, but they are not yet interoperable. Challenge: The breadth of agriculture and data regulatory organisations increases the transaction and interoperability cost, and the data can remain fragmented.

Policy priorities

Kenya is a leading scaling ecosystem, often described as a test bed for AgTech innovation across the continent. The primary policy focus must be to elevate the data ecosystem from general accessibility to an advanced, AI-ready data infrastructure, and to mature the funding landscape to support sustainable growth.

- **Policy priority 1: Accelerate national-level open data.** The national data institutions should coordinate an initiative focused on the aggregation, harmonisation, and curation of existing public and research data to create high-quality datasets for the agricultural sector. This includes standardising data cleaning and labelling processes to make the data usable to the final users.
- **Policy priority 2: Public-private data partnerships.** Kenya innovative Agtech companies have innovative products and aggregated datasets that can make government practices more efficient and close down some existing data gaps. More PPP should be facilitated through strengthened data IP and and anonymised data sharing agreements.
- **Ecosystem priority 1: Diversify and localise AgTech funding.** DFIs and impact investors often make up the majority of the AgTech funding landscape. Local commercial funders and debt providers need to be mobilised to localise and diversify the sources of funding. Some policy incentives for localising investment include tax breaks for local funders and creating risk-sharing facilities for local banks that will provide working loans to AgTechs.

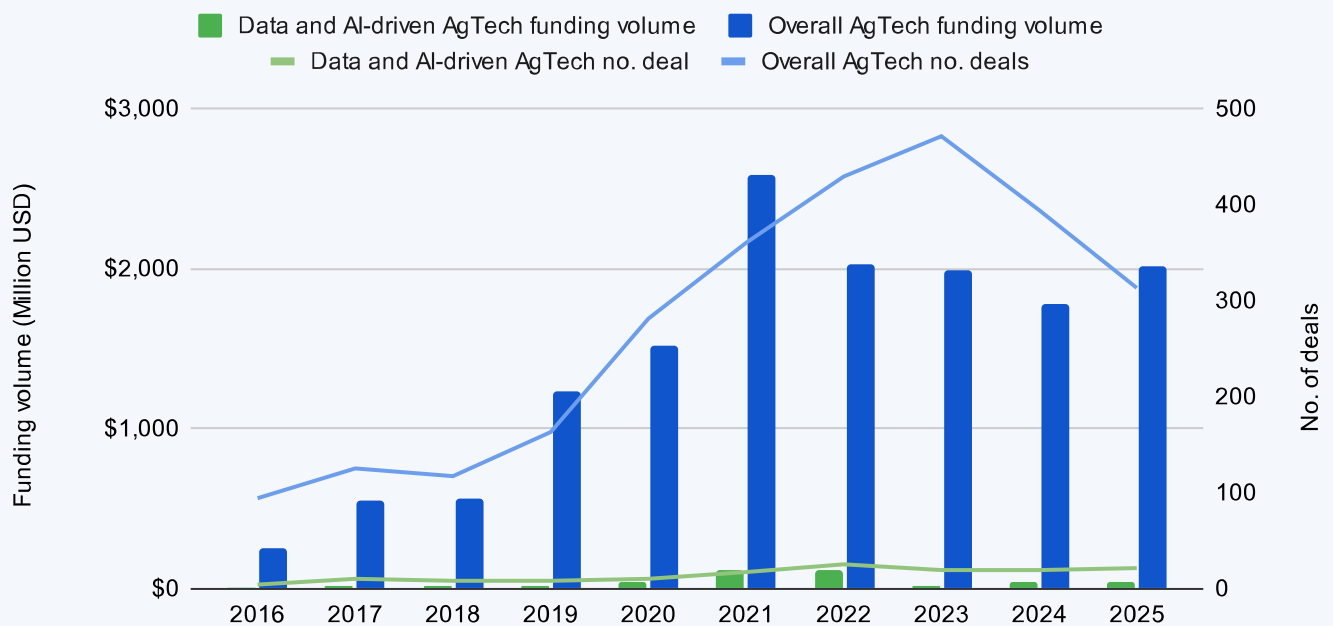
Measurable outcomes:

- Percentage of target data points aggregated and standardised amongst national agriculture datasets.
- The number of data-sharing agreements established between public and private organisations.
- Percentage increase in the number of local investors making disclosed investments in AgTech.



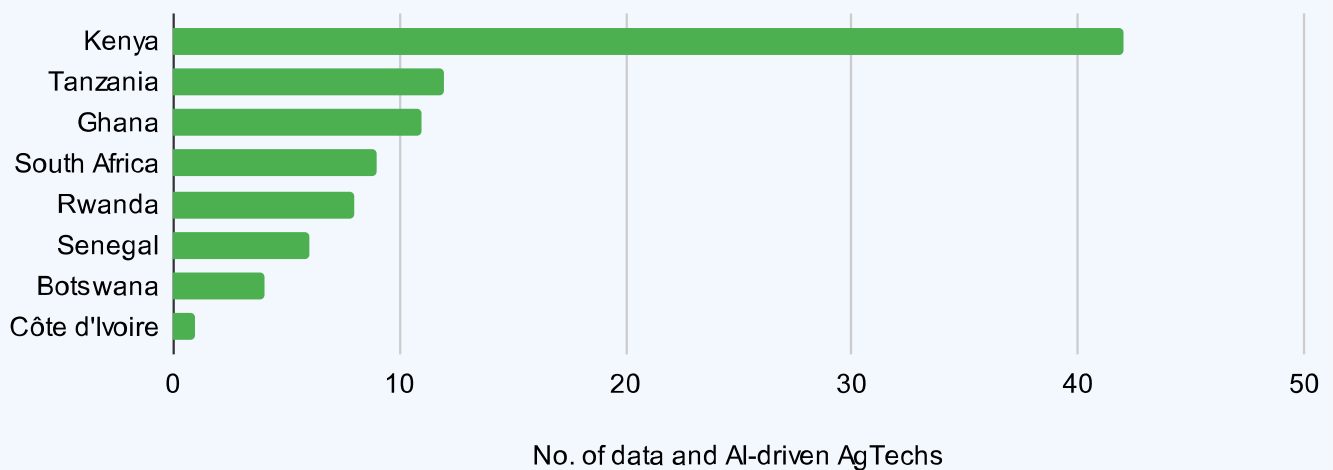
Innovation ecosystem

Figure A1. Overall funding flow in the AgTech sector



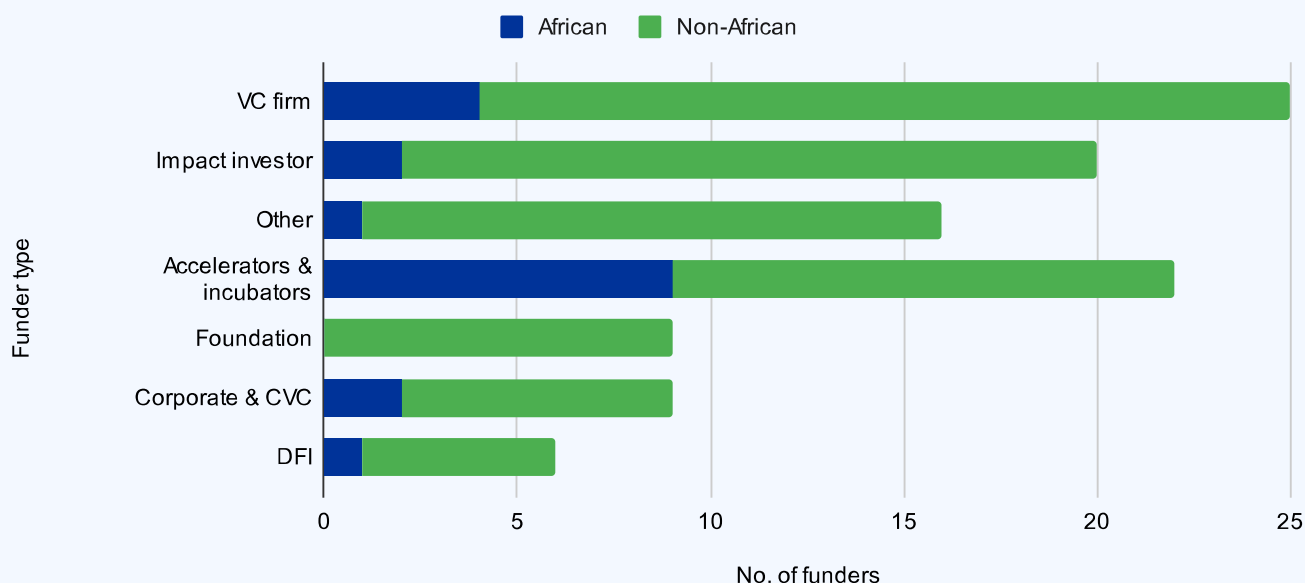
Source: Briter Intelligence and AgBase (2026)

Figure A2. Countries by the number of data and AI-driven AgTechs



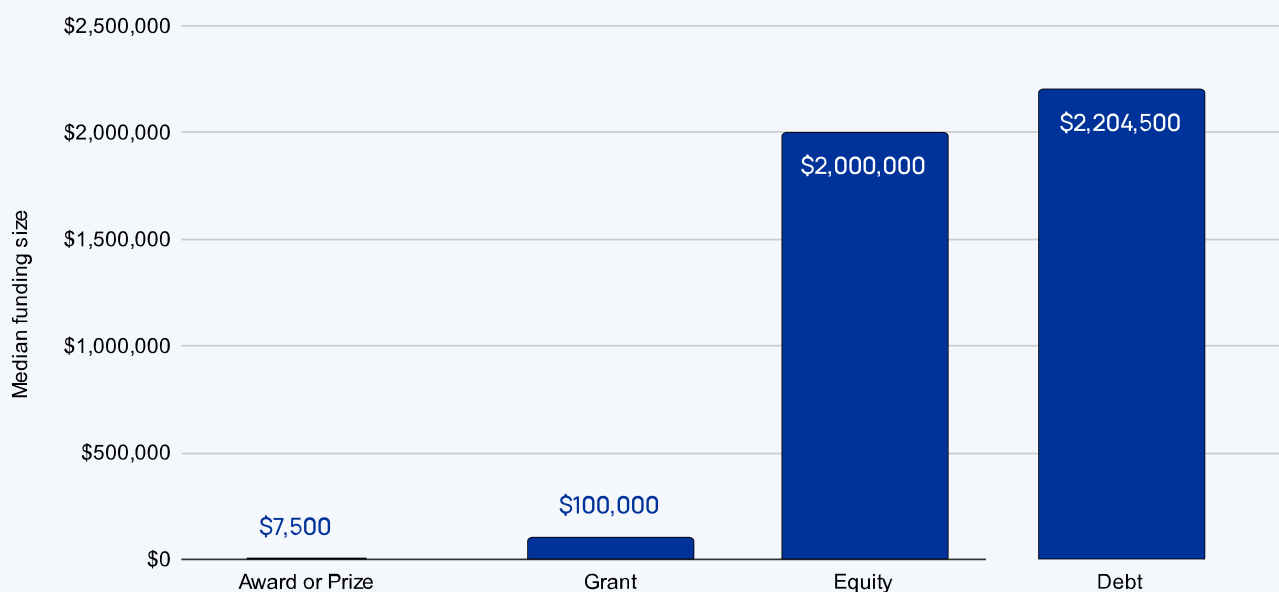
Source: Briter Intelligence and AgBase (2026)

Figure A3: No. of African and non-African investors by type



Source: Briter Intelligence and AgBase (2026)

Figure A4: Median size of funding by instrument



Source: Briter Intelligence and AgBase (2026)

Box A1. Rethinking accelerator models to equip data and AI-driven startups

Disraptor is an accelerator helping startups scale through engineering scalability in cloud, data, and AI, based in South Africa. Disraptor equips companies with scalable digital infrastructure using their specialised AI and data engineering to build data pipelines, cloud and DevOps engineering, and technical due diligence. Disraptor offers a 6-month AI Readiness Accelerator, an engineering-led programme focused on making

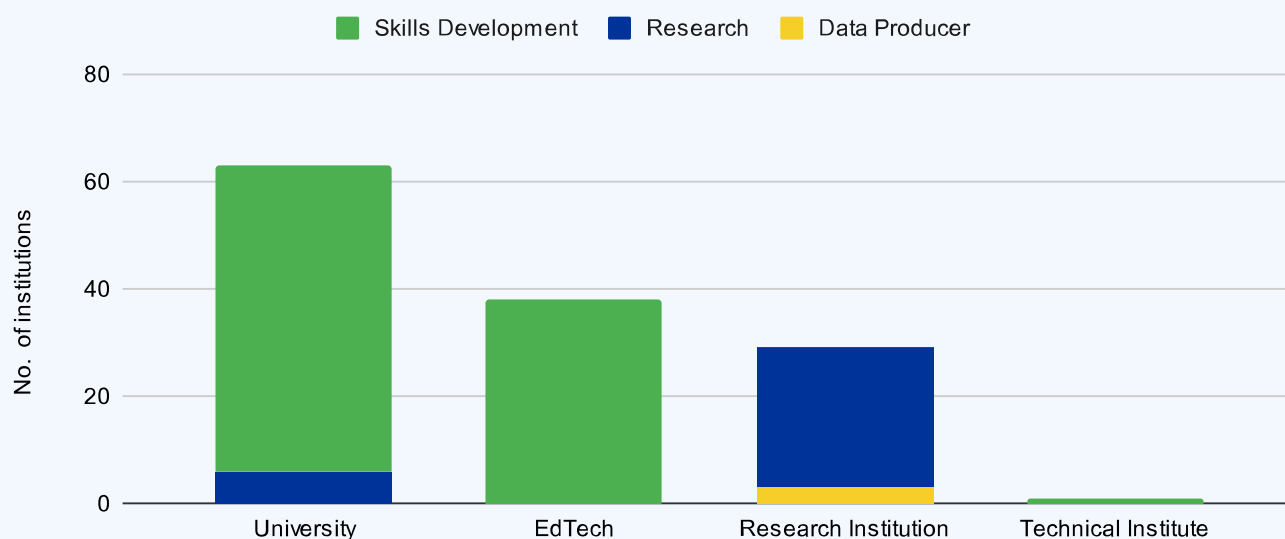
startups technically robust for scale and prepared for due diligence processes. The AI Readiness Accelerator provides bespoke services to startups through a structured delivery framework that covers cloud infrastructure and data platforms, with the Disraptor team embedding its engineers in each startup.

Disraptor also hosts the AI Founder Bootcamp to train non-technical or early-stage entrepreneurs to build, launch, and test their minimum viable products (MVPs) themselves. The bootcamp provides weekly practical sessions on AI, no-code, databases, and product validation, along with business-oriented concepts to build an MVP for traction. Entrepreneurs build their MVP through the bootcamp by leveraging a structured process to quickly build, prototype, and validate with a customer within three months.

Disraptor built these programmes as they noticed that demonstrating resilient data capabilities is increasingly important for scaling and fundraising, but many early-stage startups don't have the technical capabilities or financial resources to develop sophisticated and scalable data infrastructure from the early stage. The programme therefore focuses on building startups' infrastructure for scalability and AI-adaptiveness, and focuses on building solid cloud infrastructure for scaling, security, resilience, and cost optimisation. On the data front, Disraptor focuses on building data platforms for analytics, AI use cases, and data centricity, which they cite as critical technical due diligence to equip entrepreneurs for their onward fundraising.

Disraptor has partnered with DFIs and corporate partners to enable their acceleration support as technical assistance, so that companies can go through the programme without dilution. Disraptor has built its technical capabilities with a strong team of cloud and DevOps engineers, data engineers, solution architects, product managers, and technical leads. Through this highly technical team and strategic partnership, Disraptor helps startups build bespoke and scalable data and AI capabilities in a short period of time.

Figure A5. Training institutions by primary roles



Source: Briter Intelligence and AgBase (2026)

Box A2. University of Botswana – building digital and research foundations before AI

The University of Botswana is a public, government-funded university that is in the early stages of transitioning from a teaching-focused institution to one that places greater emphasis on research, innovation, and commercialisation. This shift began around 2021–2022 and aligns with national ambitions to diversify Botswana's economy beyond resource-intensive sectors. In recent years, the university has developed a five-year research and innovation strategy aligned with WIPO guidelines, expanded its research and innovation department, and introduced multidisciplinary research clusters to strengthen collaboration across faculties. Priority research areas include agriculture, energy, climate, and health.

As part of its innovation infrastructure, the University of Botswana hosts a UniPod funded through UNDP's Timbuktu Fund. The UniPod provides physical space and equipment for experimentation and applied innovation, while the university is responsible for long-term operational sustainability. Activities hosted through the UniPod, such as pitch competitions and maker-space initiatives, have supported early-stage applied innovations, including agritech solutions responding to local agricultural challenges.

While there is growing interest in AI, the university's experience highlights that AI readiness is constrained by limited digital foundations. Core systems and infrastructure remain outdated, including legacy hardware, old software licences, and the absence of basic digital administrative tools. At the time of the interview, the university did not have an institution-wide framework for data collection, data management, or data governance.

As a result, data and AI are not yet practical focus areas. Instead, the university is prioritising digital transition and institutional readiness. This includes developing IP and AI-related policies, clarifying ownership and income-sharing arrangements for research outputs, and building staff capacity in research management, publishing, and grant writing.

Cultural and structural factors also shape this transition. Undergraduate education and academic salaries are largely government-funded, and the public sector remains the main employer. This has contributed to a limited entrepreneurial culture within the university and resistance to equity-sharing and commercialisation models. While the university has recently filed its first patent, technology transfer and startup engagement remain nascent.

Policy landscape

Cross-border and open data policies

Alongside this, the African Continental Free Trade Area (AfCFTA) Secretariat launched the ADAPT initiative and the Trade Logistics Information Pipeline (TLIP), which produces a blockchain-enabled digital paperwork to cross-border merchants that allows different country administrations to verify information, logistics and goods trade. These flagship initiatives build use cases of cross-border data sharing for agricultural goods that move across borders, and the outcomes of these initiatives should be applied to developing a broader agriculture data sharing initiative across borders. As many of these frameworks and initiatives have emerged in the last few years, the implementation outcome and efficacy of the regulatory efforts need to be monitored in the upcoming years.

Box A3. Cross-border data challenge: When data can't travel, startups can't scale

Fragmented regulatory and infrastructure environments, particularly disparate national data protection regimes, are the primary barrier preventing African data and AI-driven startups from scaling across borders. This forces companies to design systems for single-country deployments, prioritising compliance over pooling data across markets for improved analytics. Continental efforts, such as the African Union and AfCFTA Digital Trade Protocol, aim to create interoperable data governance frameworks, but until then, data and AI-driven startups are likely to remain nationally siloed.

Figure A6. Summary of policy initiatives by key countries

Country	AI Strategies	Data Privacy	Data Privacy
South Africa	2026 AI Execution Strategy: Pivots industry from training to large-scale inference; focuses on agentic AI in mining, automotive, and finance.	Protection of Personal Information Act (POPIA); 2026 guidance note on automated decision-making requires explanations for AI decisions.	Patents Amendment Bill (2025-2026): Introduces Substantive Search and Examination (SSE) and clarifies AI-generated inventorship.
Kenya	National AI Strategy (2025-2030): Prioritises AI in agriculture and health-care; establishes a national research hub for governance norms.	Data Protection Act (2019); 2026 Audit Framework mandates DPIAs for AI firms processing biometric or high-risk data.	Robotics and AI Society Bill (2025): Establishes regulatory body for human-in-the-loop standards; restricts IP to innovations with human effort.
Ghana	National AI Strategy (2023-2033): Focuses on inclusive growth, public service delivery, and the smart electricity grid.	Data Protection Bill (2025): Updates 2012 Act to include "Right to Human Intervention" for AI decisions in banking and healthcare.	Emerging Technologies Bill (2025): Unified framework for AI/Blockchain; creates certification for priority IP processing for local developers.
Rwanda	National AI Policy (2023-2025): Focuses on a world-class AI university ecosystem and regional cloud infrastructure.	2026 certification programme: AI companies must prove privacy-by-design to access national data centre infrastructure.	Regulatory sandboxes for AI: Allows startups to test proprietary algorithms under temporary, flexible IP enforcement.
Botswana	SmartBots (4IR) Strategy: Integrates AI into digital transformation; focuses on knowledge sharing and digitising indigenous systems.	Data Protection Act (2024): Fully operational in 2025; strict clauses against automated profiling without explicit consent.	Industrial Property Act (Updated 2025): Mandates licensing and benefit-sharing for AI trained on traditional or indigenous knowledge.
Tanzania	National AI Strategy (2025): Informed by UNESCO Readiness Assessment; focuses on reducing the digital divide through skills training.	Personal Data Protection Act (2022/23): Strict data localisation; models trained on Tanzanian data must undergo PDPC security audits.	Copyright Amendment Bill (2025): Clarifies fair use for AI training; stipulates that purely AI-generated outputs cannot claim legal authorship.
Côte d'Ivoire	Stratégie Nationale de l'IA (SNIA)(2030): Focuses on investment and modernisation of agriculture and cultural digital archiving.	National Register of Data Protection Officers (DPOs): DPOs are legally responsible for algorithmic transparency in their organisations.	"Safe AI" Labelling Decree (2025): Integrates IP with ethical labelling; grants legal recourse against unauthorised

Senegal

Digital Senegal 2035 Strategy: Focuses on supporting local startups and enhancing the digital economy's competitiveness.

National Data Strategy: Requires AI providers to guarantee sensitive personal data remains within national borders during training.

Startup Act (Revised 2025): Innovation labels provide automated IP registration assistance and international patent subsidies.

Box A4. Data interoperability in practice

Interoperability is the ability of different systems to "speak" to each other and use exchanged information without manual intervention. It functions through four distinct layers, which can be applied in the AgTech sector as follows:

1. Technical – how is the information collected? There are physical infrastructure and protocols that move bits of data from the farm to the cloud.
2. Syntactic – how is the information formatted? The standardised format of the collected data, so that instead of random text, systems use JSON or XML so the receiving software knows where the "Date" field ends and the "Yield" field begins.
3. Semantic – what does the data mean? There are standardised vocabularies to ensure that when System A says "Maize," System B doesn't interpret it as "Corn" or a different crop variety. In the AgTech world, systems like AgroVoc help with standardisation.
4. Organisational – how does the data move around, and by whom? The legal and business rules of AgTechs define who has permission to see the data and ensure workflows between a startup and a bank are aligned. For example, the ODPC Data Sharing Code of Kenya defines these rules and parameters.

Diagnostic framework

Below, the three dimensions of the diagnostic framework are broken down by indicator and mapped against each country in the framework. Each country is given a score between 0 (no data), 1 (low activity/performance), 2 (below median but noticeable activity/performance), 3 (above median activity/performance), and 4 (high activity/performance).

1. Ecosystem development: This category assesses the state of the AgTech innovation ecosystem development by sizing the AgTech market, the support mechanisms existing in the ecosystem, and the funding into the ecosystem. The data collected for this category are mainly from the Briter Intelligence data, which follows activities of 250+ funded AgTechs and 200+ funders and ESOs across 8 focus markets in the last 10 years. Under this assessment category, each countries are scored between 0 (no data) to 4 (higher rating) based on the following areas:

- **AgTech sector market size**, measured by % funding size into agtech sector
- **AgTech innovation size**, measured by the proportion of AgTechs in the broader ecosystem
- **Ecosystem support availability**, measured by the number of AI and data specific ESO programmes

operational in the last 5 years

- **Data application within AgTech sector**, measured by % of data-driven agtechs
- **AI application within AgTech sector**, measured by % of AI-driven agtechs
- **Investment into AI and data-driven AgTechs**, measured by % of funding count into AI and data-driven AgTechs
- **Gender inclusion within AgTech**, measured by % of AgTechs founded by women

Figure A7: Ecosystem development diagnostic framework assessment

Ecosystem development	Botswana	Côte d'Ivoire	Ghana	Kenya	Rwanda	Senegal	South Africa	Tanzania
AgTech sector market size	0	1	3	2	1	2	1	4
AgTech innovation size	4	2	13	2	3	4	1	4
Ecosystem support	2	1	2	4	3	2	3	2
Data application within AgTech sector	0	4	4	4	4	4	4	2
AI application within the AgTech sector	0	1	1	2	3	1	2	1
Investment	0	0	1	2	3	2	2	1
Gender inclusion within agtech	0	1	2	2	1	3	1	2
Total out of 28	6	10	16	18	18	18	14	16

2. Macro environment: This category assesses the overall economic and agricultural conditions that serve as the baseline for the market conditions and agricultural sector in which AI and data-driven AgTechs operate. The data collected in this category are the latest open data on macroeconomic activities from the World Bank, FAO, and SME Finance Forum. Macro environment is assessed and scored between 0 (no data) and 4 (higher rating) by the following variables:

- **Agriculture market size** as % of GDP
- Size of the population **employed in the agriculture sector**
- **Internet infrastructure**, measured by the availability of secure internet servers per million people
- **Internet penetration on mobile devices**, measured by cellular subscription rate
- **Entrepreneurship enablement**, measured by MSMEs per 1000 people

Figure A8: Macro environment diagnostic framework assessment

Ecosystem development	Botswana	Côte d'Ivoire	Ghana	Kenya	Namibia	Rwanda	Senegal	South Africa
Agriculture market size	0	1	3	2	1	2	1	4
Agriculture employment size	4	2	13	2	3	4	1	4
Internet infrastructure	2	1	2	4	3	2	3	2
Internet usage	4	4	2	3	1	1	3	4
Entrepreneurship Enablement	1	1	3	2	0	1	2	1
Total out of 20	10	14	13	14	9	11	12	14

3. Enabling environment: This category measures regulatory and data infrastructure that enables data and AI technology development across the broader innovation ecosystem as well as more specifically in the agriculture sector. The analysis in this category is drawn from data sources such as the government in each country, World Bank, desk research in each data source, UNESCO, and Open Data Inventory Network. 8 countries in the study are assessed and scored from 0 (no data) to 4 (higher rating) against the global average across the following areas:

- **Data policy**, assessed by the presence and implementation of the national open-data policy
- **Data governance**, assessed by the presence and governance of national data protection and governance laws
- **Number of open datasets** about the agriculture sector
- **Standard & interoperability**, measured by open datasets with API and other common data formats
- **Gender inclusion for talent development**, measured by % women in STEM education
- **Availability of open data**, measured by the statistical matching to international open data standards provided by the Open Data Inventory

Figure A9: Enabling environment diagnostic framework assessment

Assessment Area	Botswana	Côte d'Ivoire	Ghana	Kenya	Rwanda	Senegal	South Africa	Tanzania
Talent pipeline	2	2	3	3	1	2	3	2
Data policy	2	2	1	3	2	2	3	3
Governance	0	0	0	0	0	0	0	0
Ag Data availability	2	3	3	4	3	4	4	3
Gender inclusion for talent development	0	0	1	1	1	0	4	0
Availability of data	2	3	2	3	3	3	2	2
Total out of 24	8	10	10	14	10	11	16	10

